

Final Predation Kernels

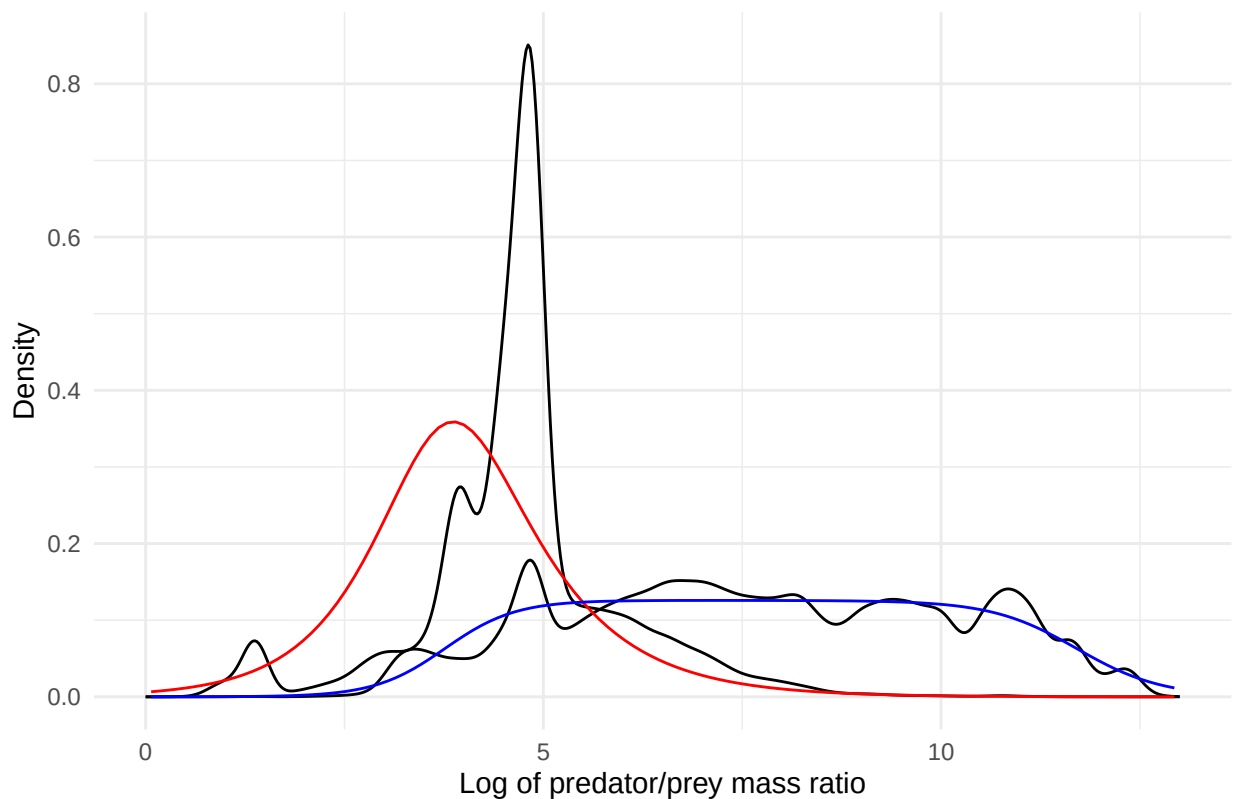
2024-07-22

Cod

[Cod PDF file](#) [Cod RMD file](#)

For Cod, the fit wasn't very good. The structure of the PPMR distributions are different compared to all other species, as it has multiple small peaks. I think this is due to the large dataset that we have for cod. However, this irregular PPMR density shape means that the shifted fit of biomass from number distribution is not very good. Additionally, as we have a lot of data, the structure of the data may be affecting the fit, as it might have many different years and regions. So if we wanted a better fit, we could subset the data for the most recent years and try again.

Cod – Fit to number, shift to diet



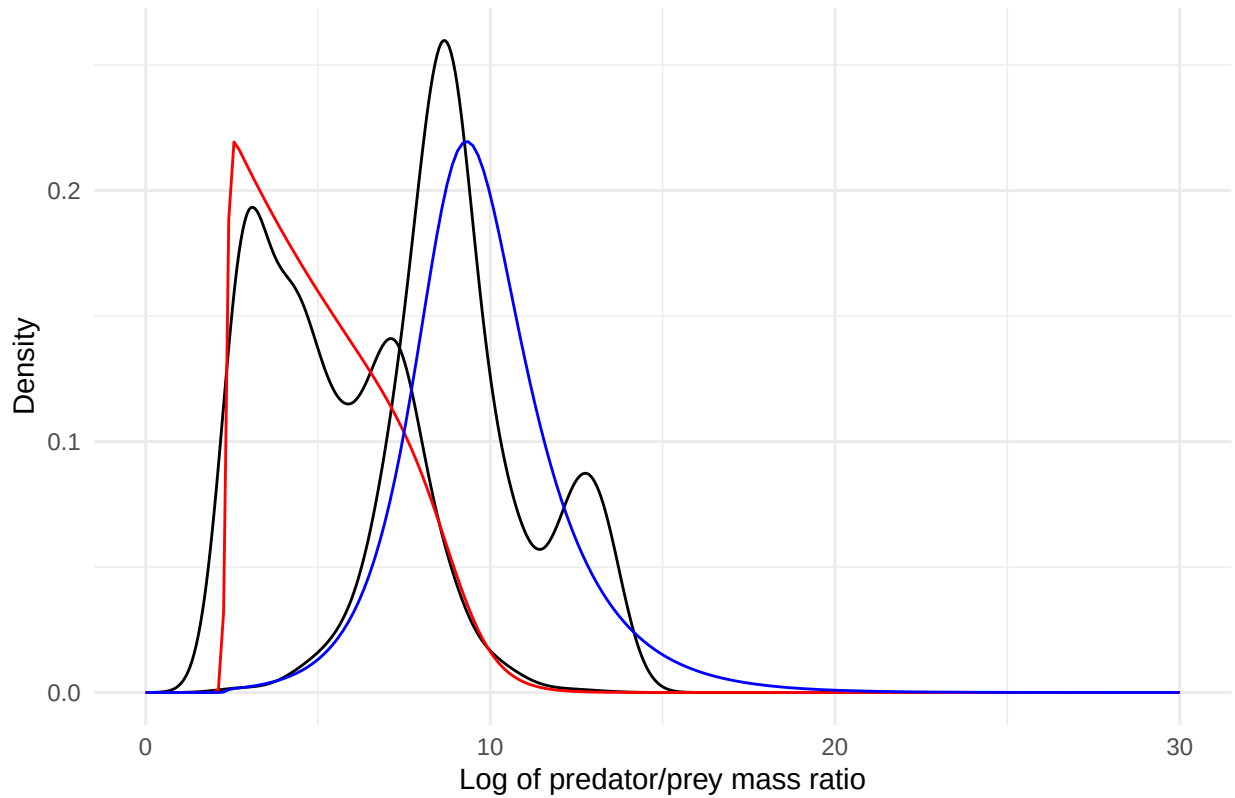
Blue Whiting

[Blue Whiting PDF file](#) [Blue Whiting RMD file](#)

The fit for Blue Whiting was very good. But it appears that there are 2 feeding modes and that they eat

very large prey (corroborated [here](#)) The feeding modes have also been described in [literature](#) and they may be due to seasonal food availability.

Blue Whiting – Fitted to Biomass, shift to number



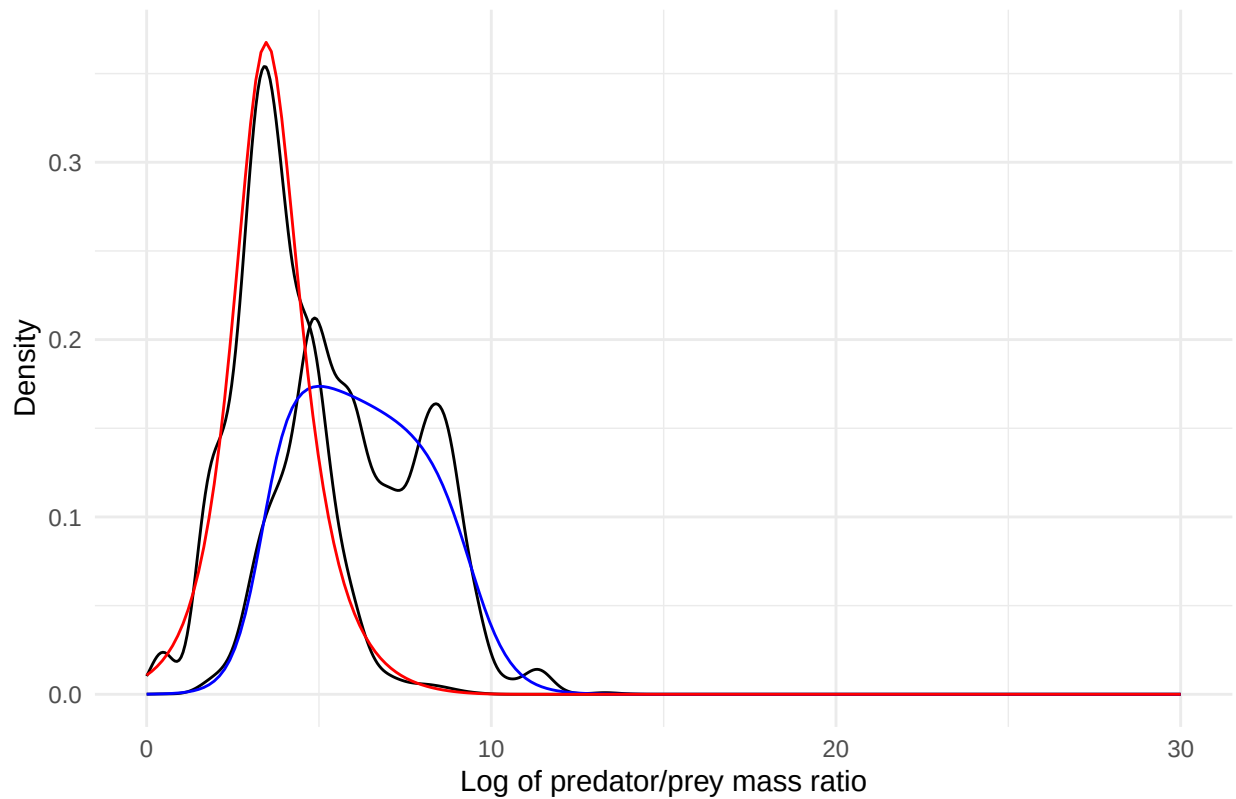
Common Dab

[Common Dab PDF file](#) [Common Dab RMD file](#)

Common Dab has a very good fit to the data. But appears to have two feeding modes - this is most likely due to two different years or areas. As the feeding ecology of dab is that it is a highly generalist [feeder](#), so two separate feeding modes would not make any sense.

(After checking the structure, there is no difference in the years sampled between the two feeding modes, and there doesn't appear to be a difference in the regions that the samples were taken from) So therefore it is likely that the appearance of the feeding modes is due to the fact that we haven't sampled enough.

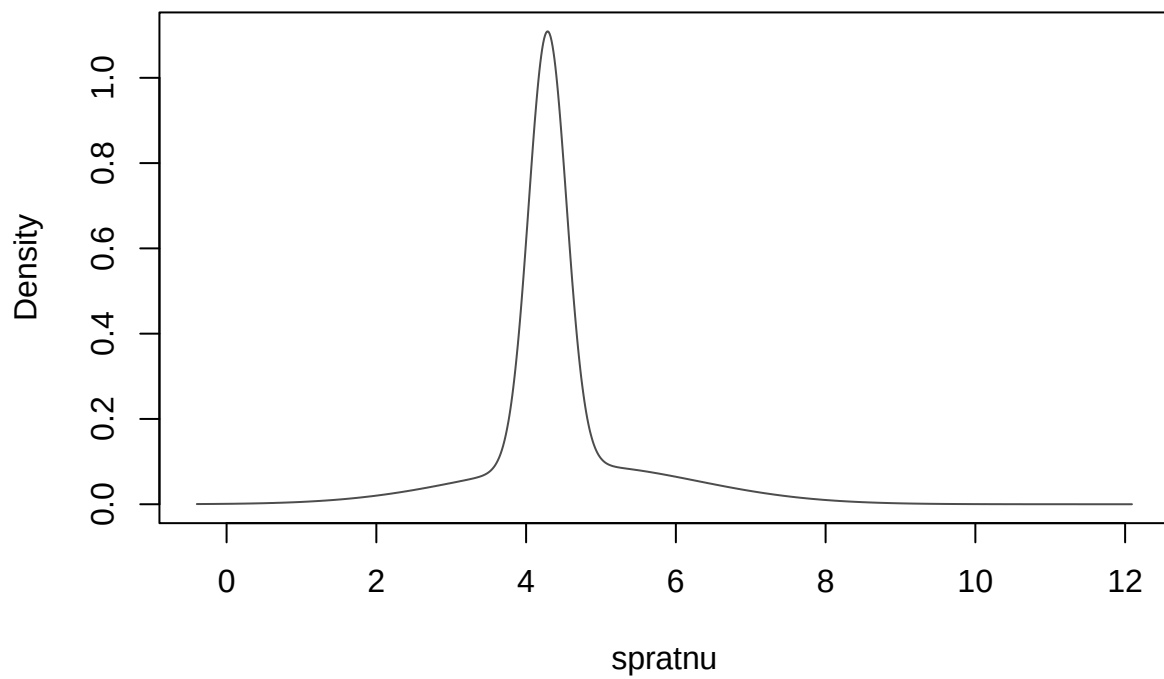
Common Dab – Fitted to number, shift to biomass



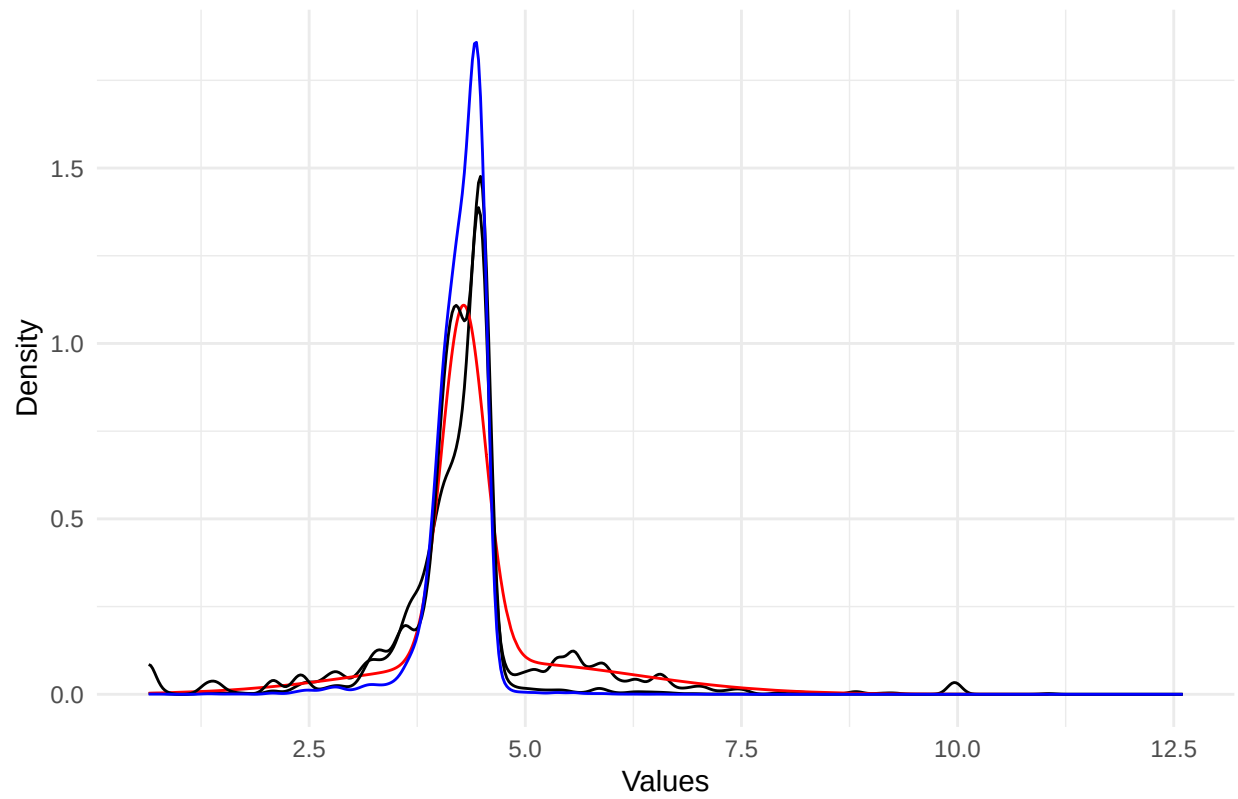
European Hake

[European Hake PDF file](#) [European Hake RMD file](#)

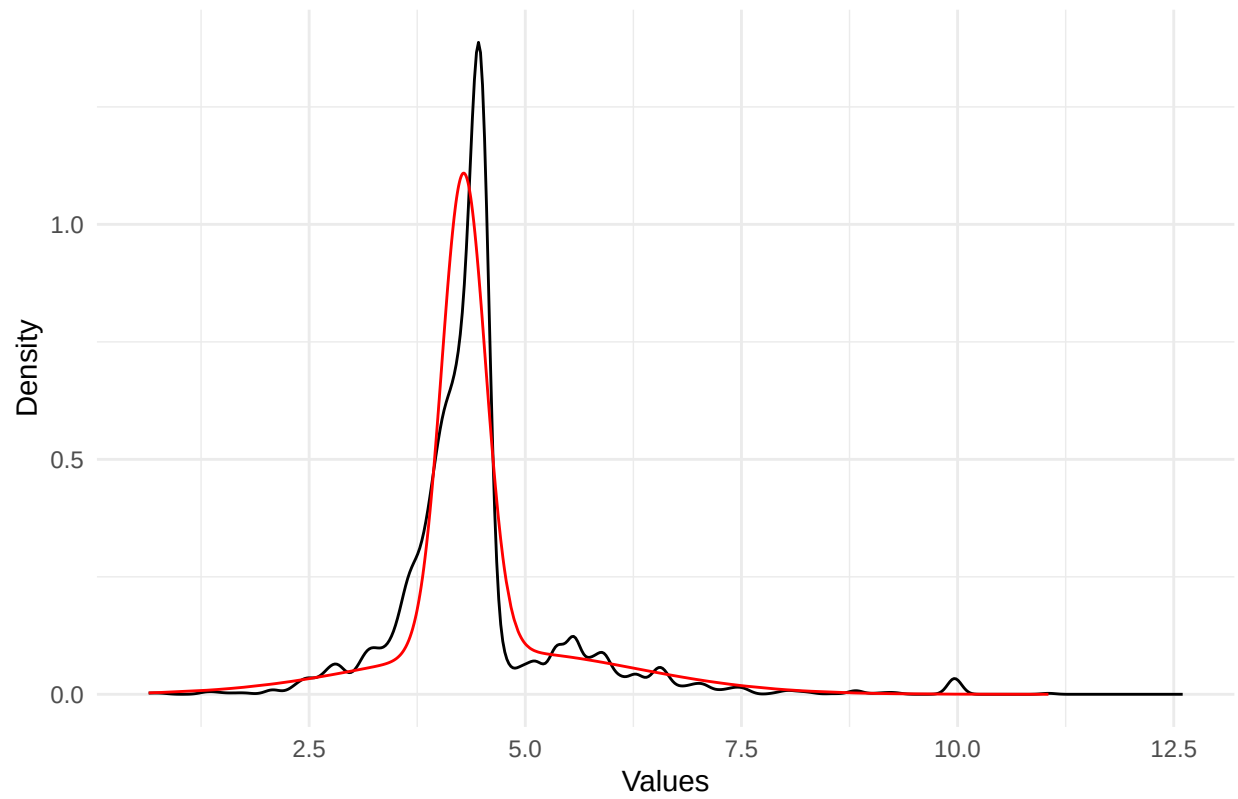
European Hake has a good fit, however, strangely, accounting for biomass does not change the density distribution. (this makes it very hard to look at the fitted functions, so they have been separated). Also strangely, it appears to be a very specialist predator, it has a single density peak and a low variance. But the [literature](#) doesn't support this fact, it says that European Hake interacts with nearly every species in the ecosystem, and eats everything. So I guess it does eat a range of species, but will only eat a certain size of species, additionally, it only eats species with a high biomass, and doesn't eat any prey that won't contribute greatly to its diet.



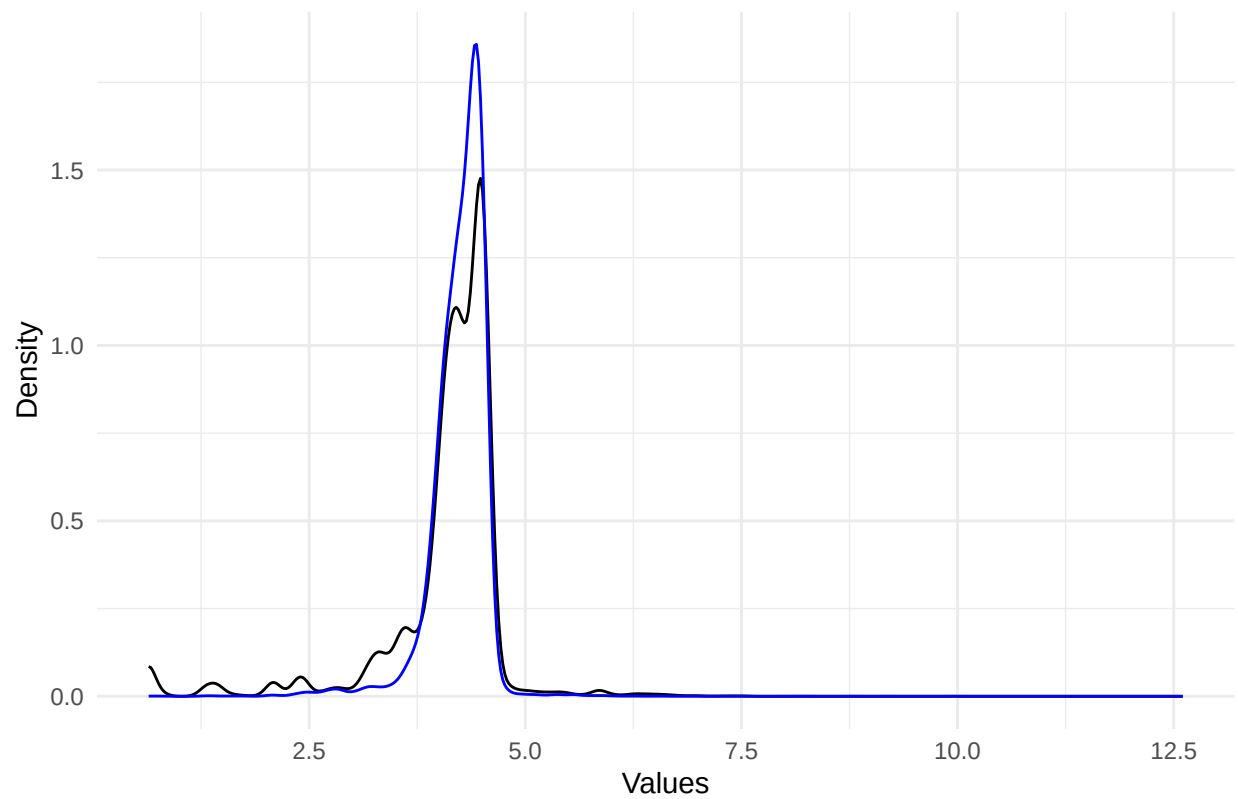
European Hake – Fitted to number, shifted to biomass



European Hake –Number Density Plot from Number Distribution



European Hake –Diet Density Plot from Number Distribution

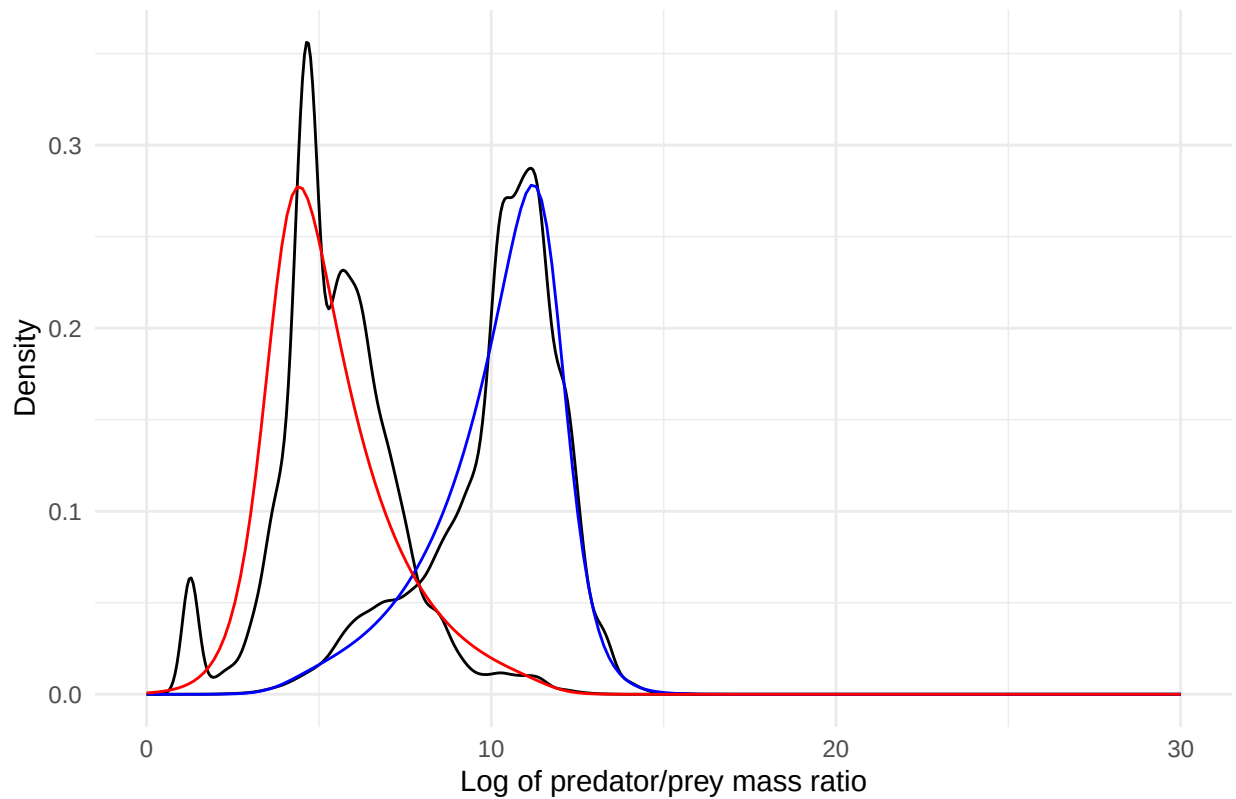


Haddock

[Haddock PDF file](#) [Haddock RMD file](#)

Haddock has a good fit. I couldn't find much on the feeding ecology of Haddock to match these results to, but Haddock has a lot of stomach contents data, so I think it is okay to believe these results.

Haddock – Fitted to number, shift to biomass

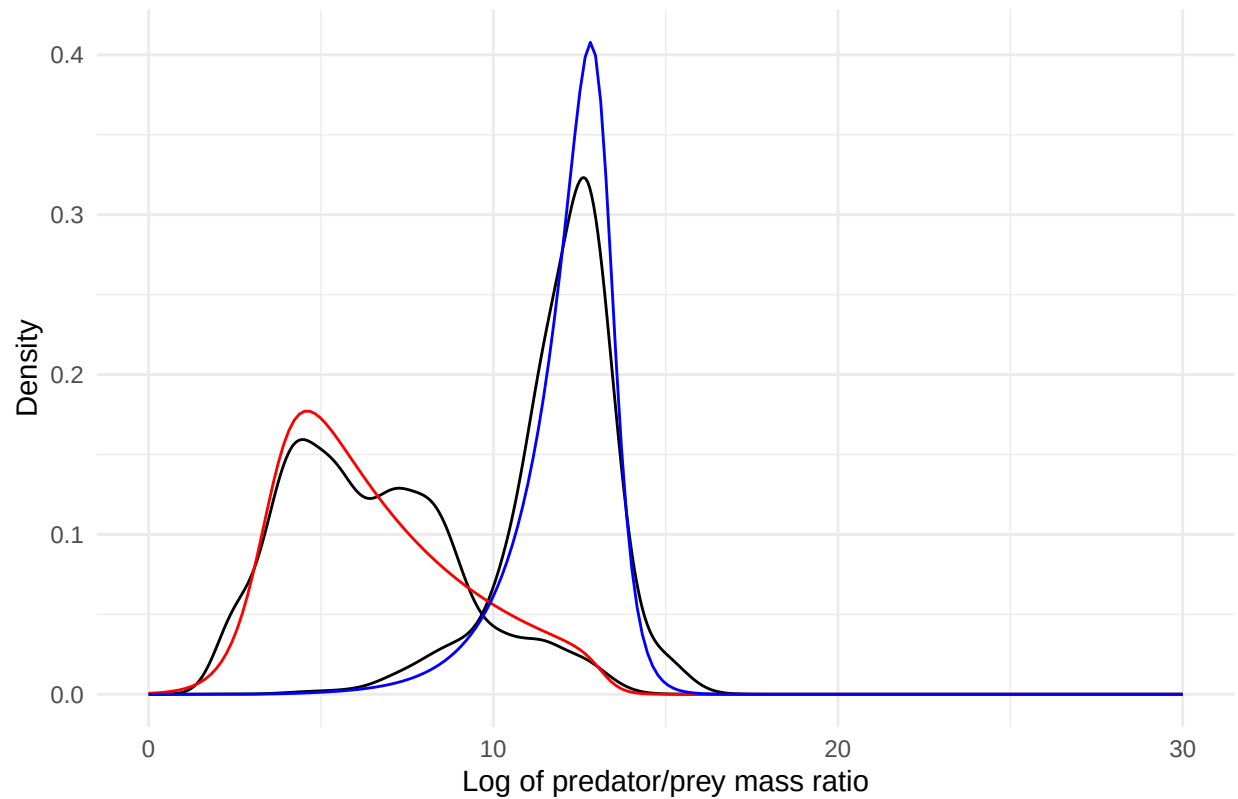


Horse Mackerel

[Horse Mackerel PDF file](#) [Horse Mackerel RMD file](#)

Horse Mackerel has a good fit. The fact that there appears to be multiple feeding modes in the biomass fit could be due to the fact that Horse Mackerel feeding behavior has been shown to be due to [environmental factors](#), and the data has been collected from multiple areas.

Horse Mackerel – Fitted to biomass, shift to number

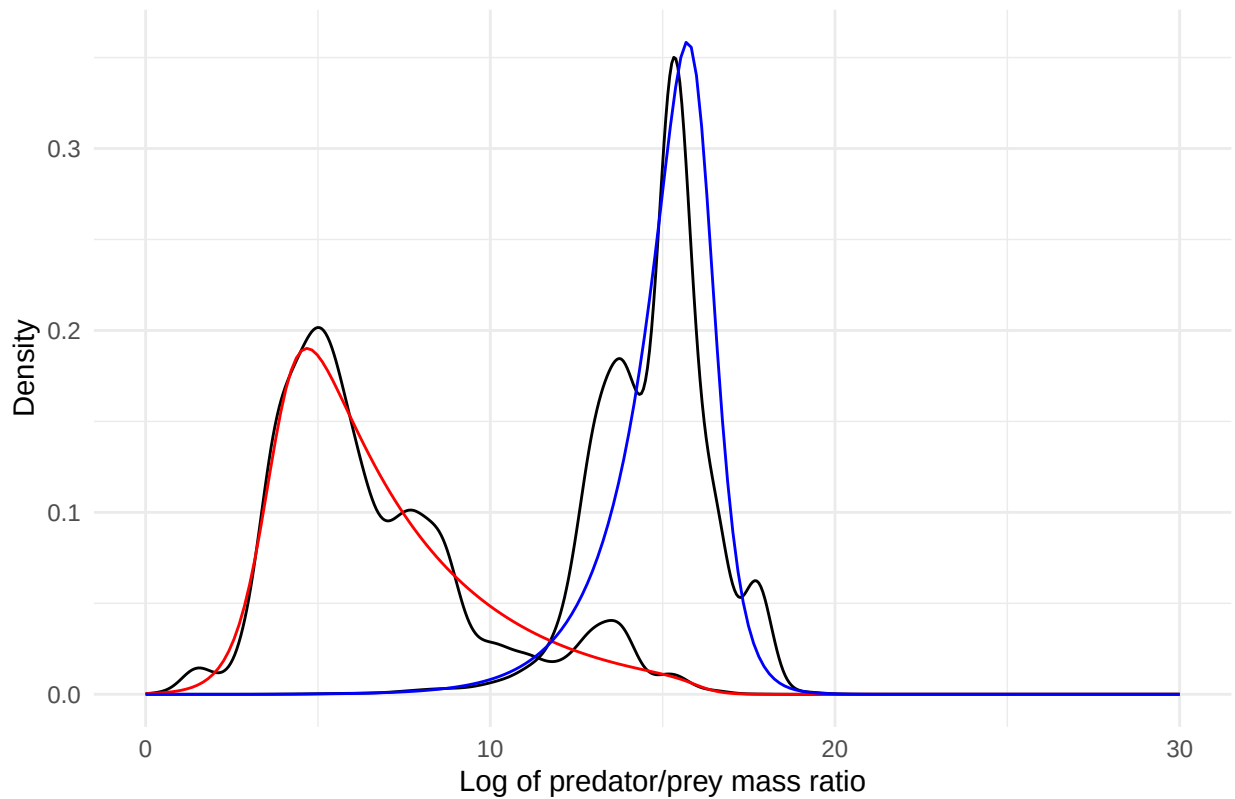


Mackerel

[Mackerel PDF file](#) [Mackerel RMD file](#)

The mackerel fit is good, and is very similar to the fit to the horse mackerel. Mackerel has a wider density function than other species, maybe due to the fact that it has seasonal migrations and also is an opportunistic predator. [Source](#)

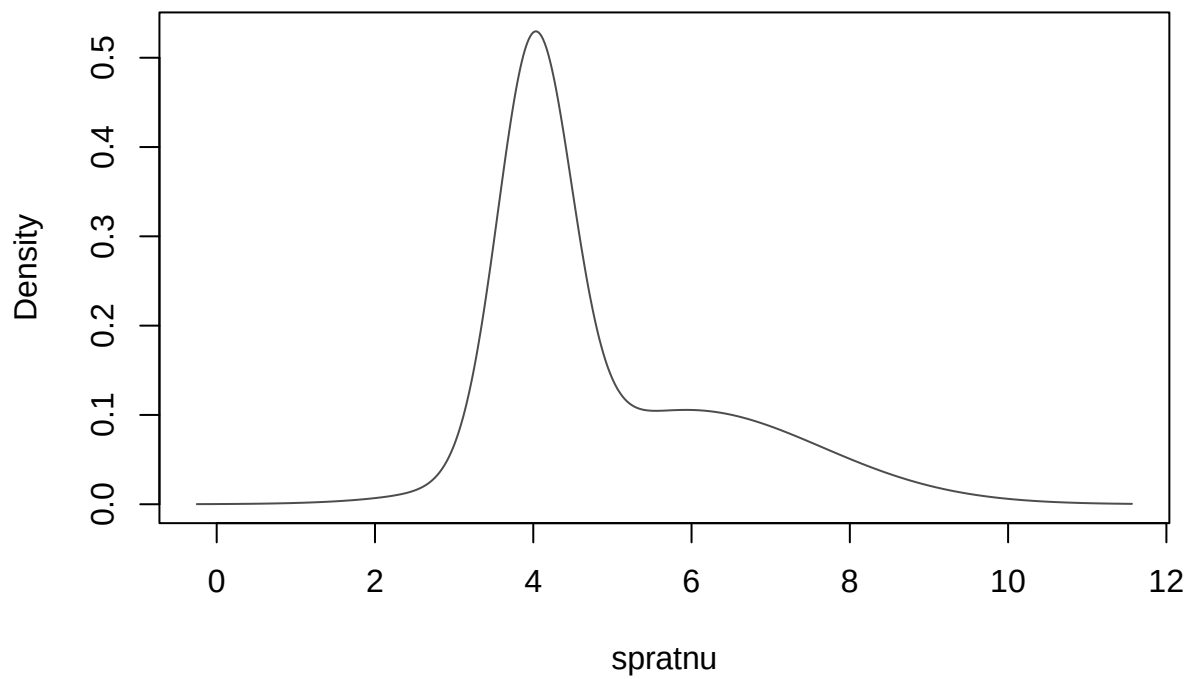
Mackerel – Fitted to biomass, shift to number



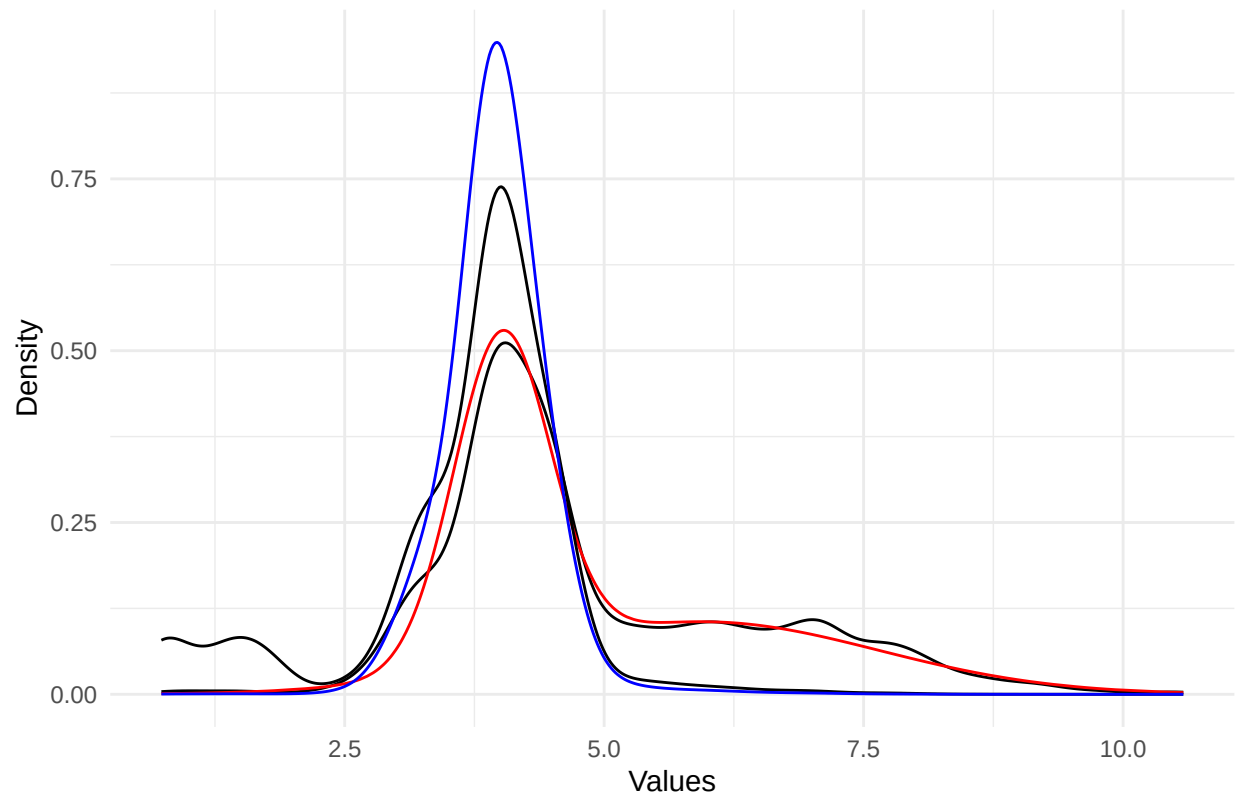
Megrim

The fit to Megrim is good I think. Megrim seems to feed primarily on items of prey that contribute highly to the diet, but there is also a large range of prey sizes that Megrim feeds on. What this could be, is that Megrim has been noted to have large seasonal variations in prey availability (this might be only because the study was in the Adriatic Sea), meaning that in winter the proportion of Megrim with empty stomachs increases. Therefore, in winter, the diet desperation might mean that they expand their prey selection kernel.

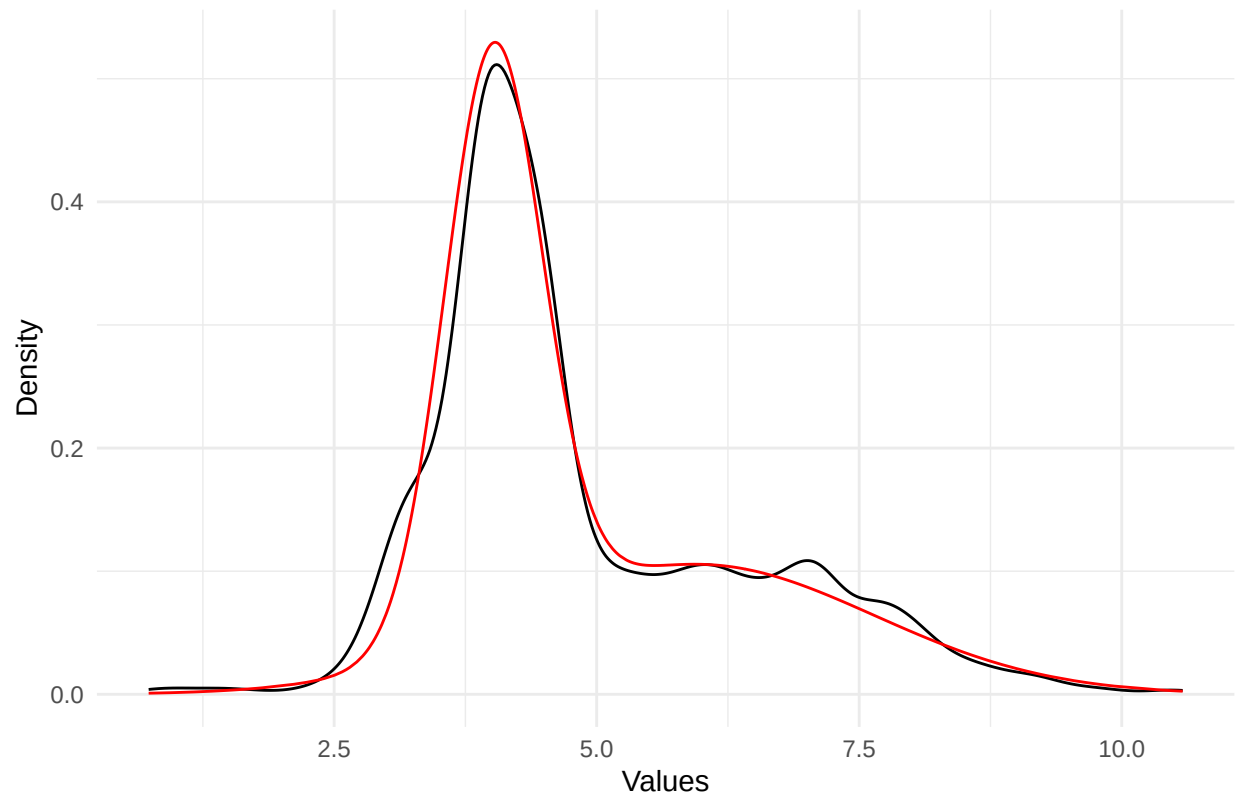
[Source](#)

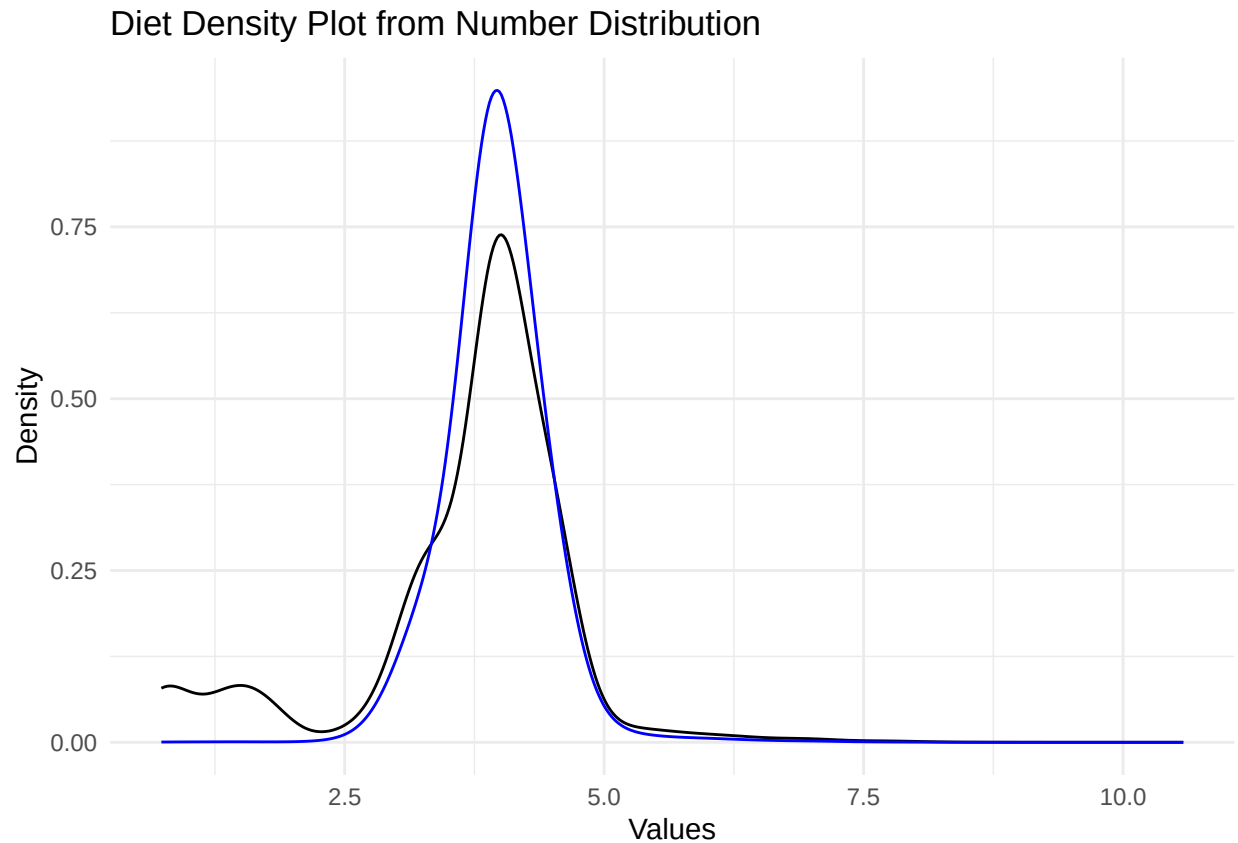


Megrim – Fitted to number, shifted to biomass



Number Density Plot from Number Distribution



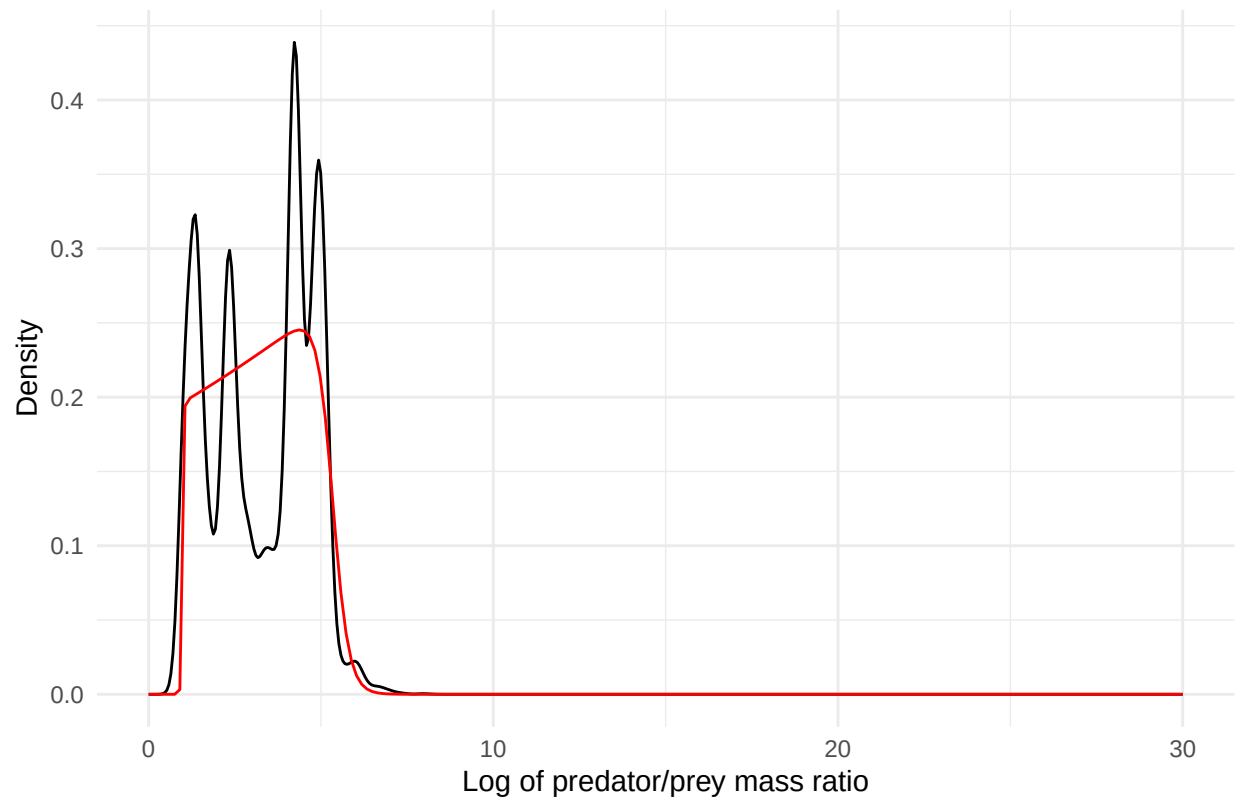


Monkfish

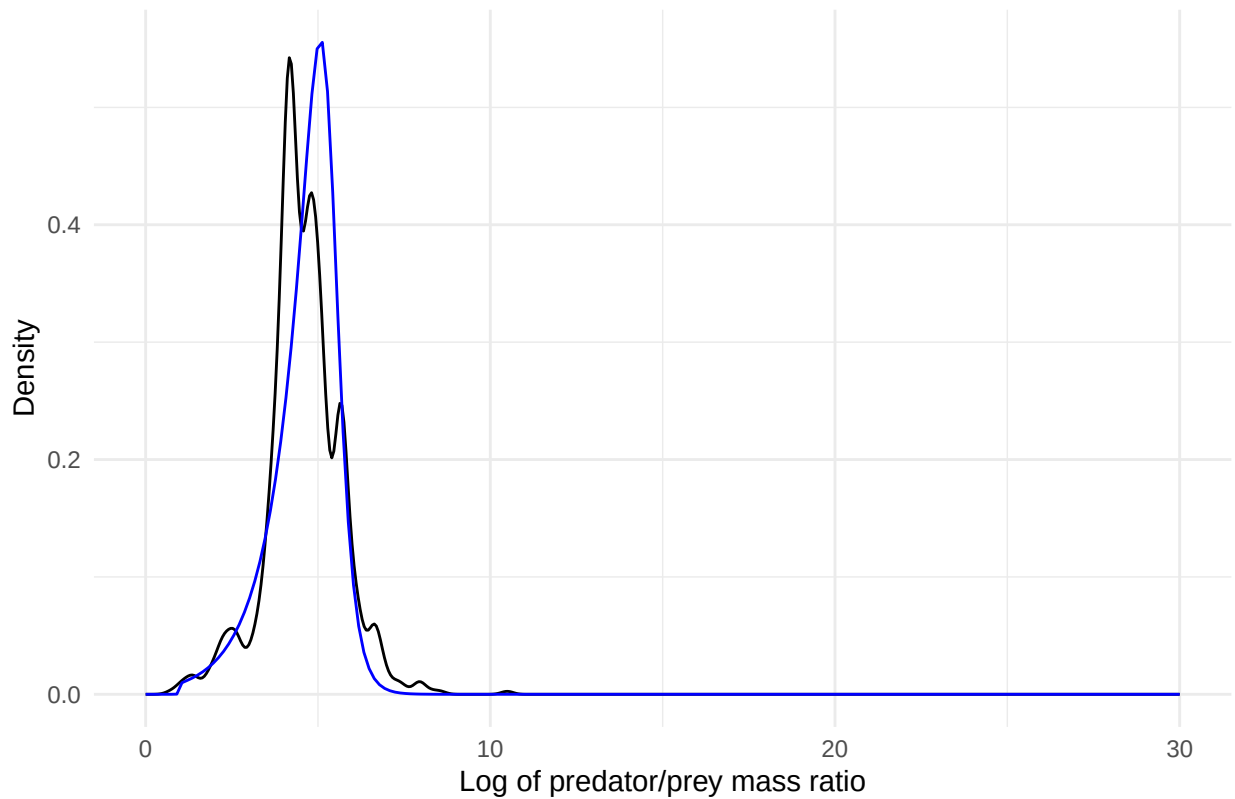
[Monkfish PDF file](#) [Monkfish RMD file](#)

Monkfish has an okay fit. It is difficult to fit to this PPMR distribution as it has many separate peaks. We do not have much data on monkfish, but it seems to be enough. Monkfish has a very large mouth so it makes sense that the PPMR is low, and the separate peaks may be because it is opportunistic and so the peaks are the only prey available.

Monkfish – Fitted biomass



Monkfish – shifted to number

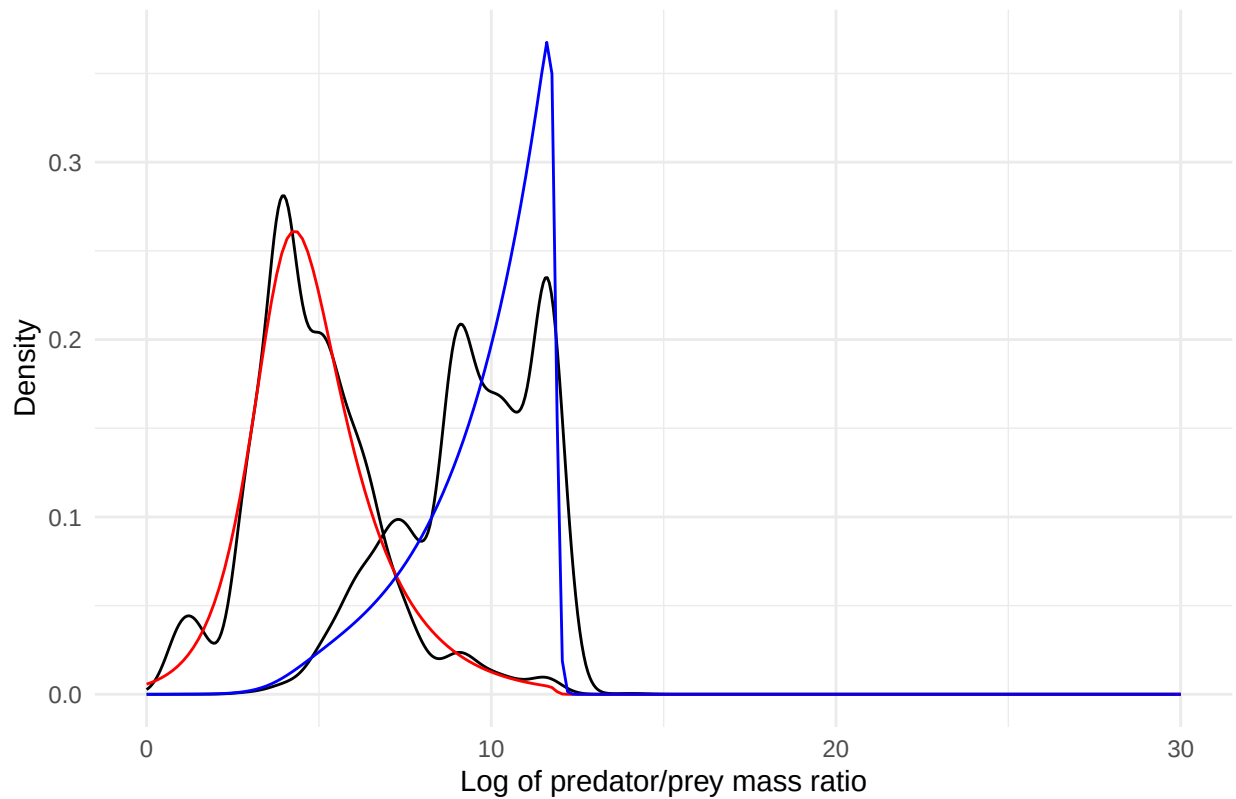


Norway pout

[Norway pout PDF file](#) [Norway pout RMD file](#)

The fit to Norway Pout is okay. The distribution for biomass and number both have many small peaks, which is probably due to the small sample size. However, it also might be due to Norway pout moving deeper in the water column as it gets bigger/older [Source](#), so therefore there could be multiple feeding modes. Also, it is a relatively small species, and feeds on larvae, plankton, and small fish. So therefore, there may be some sampling bias as these items digest fast and are hard to study. Regardless, the fit is good enough.

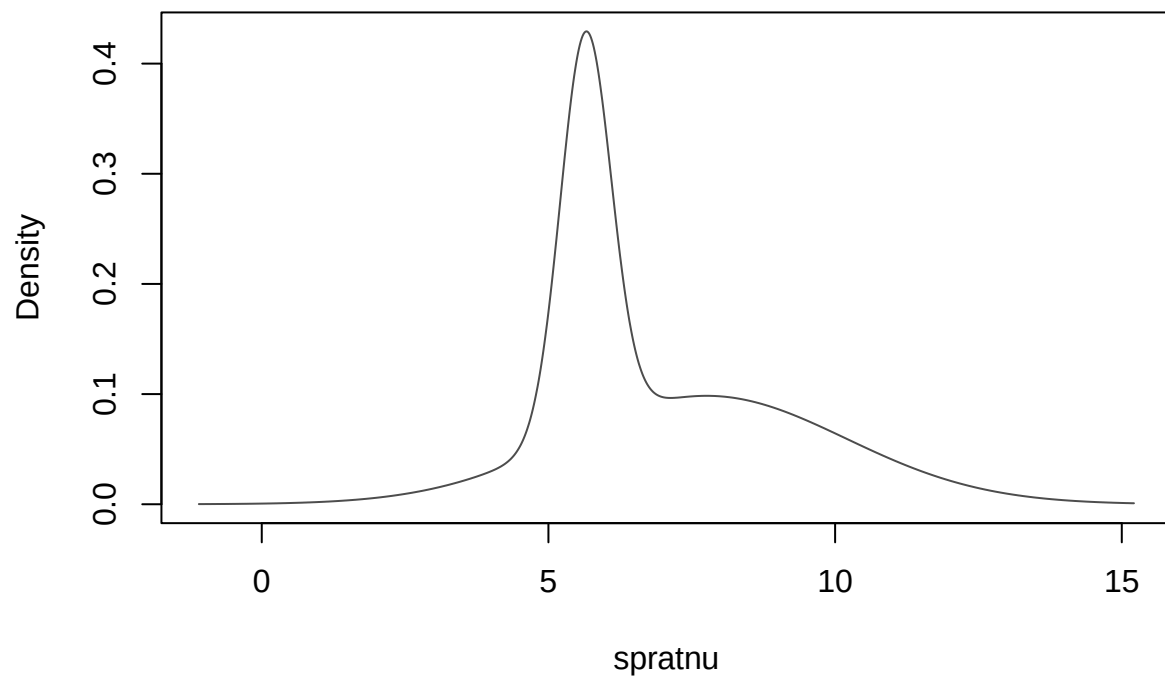
Norway Pout – Fitted to biomass, shift to number

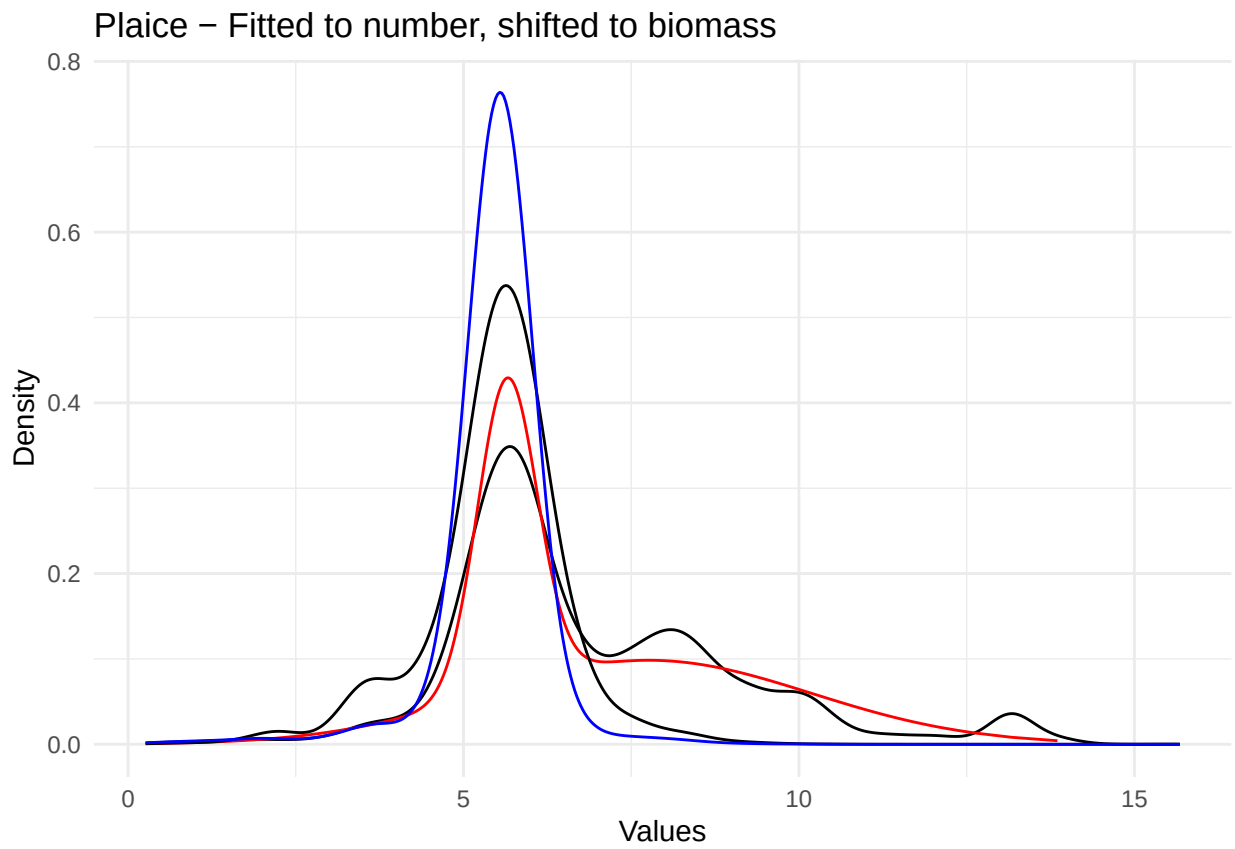


Plaice

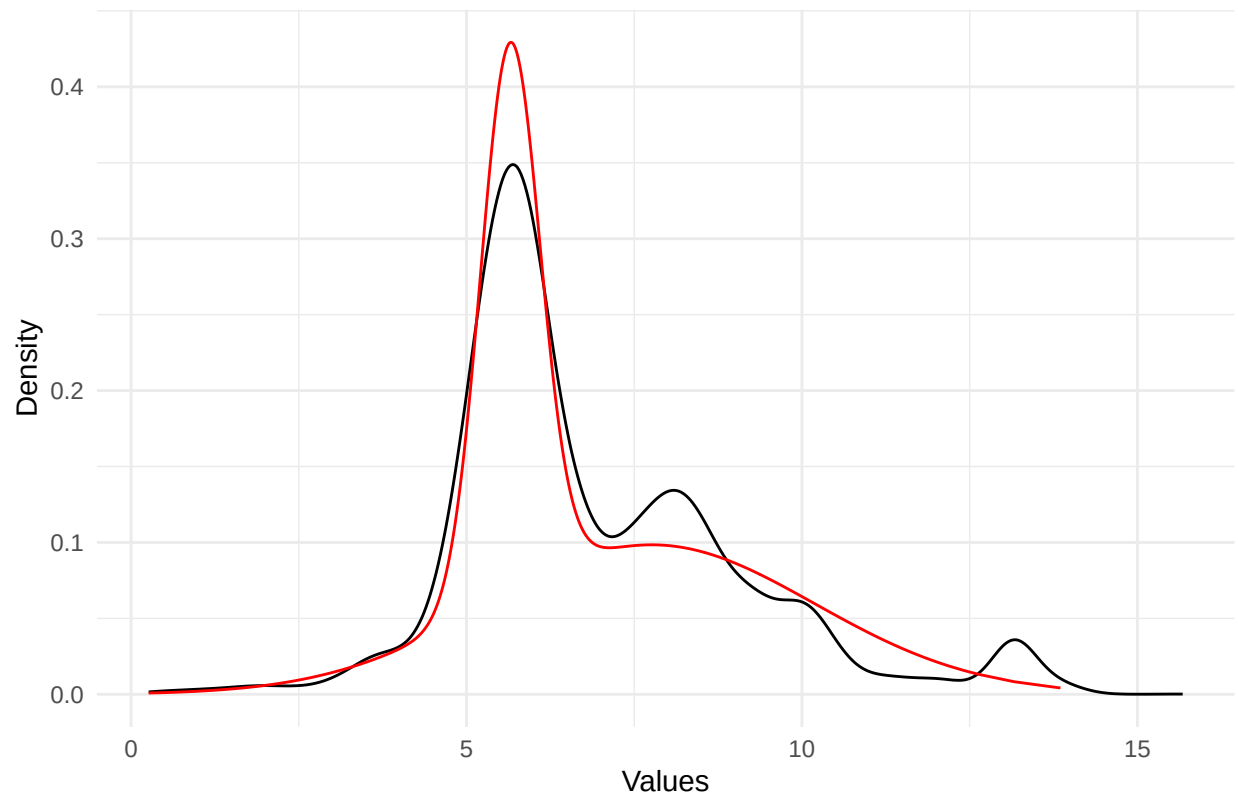
[Plaice PDF file](#) [Plaice RMD file](#)

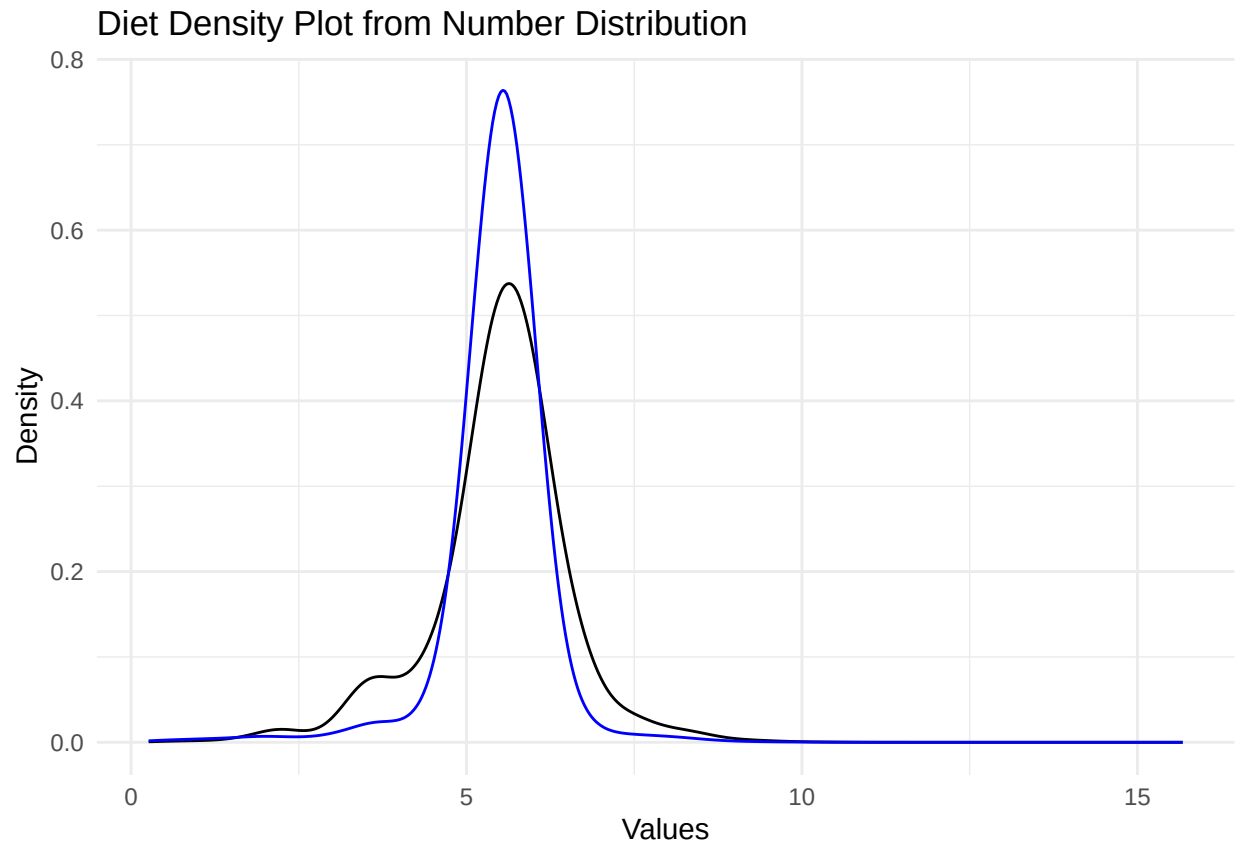
Plaice has a pretty good fit for how hard the data is to fit to. The fit is also very similar to Megrin, but not to the Common Dab, even if they are all flatfish (I think) (This makes sense, as the Common Dab and Plaice have been shown not to compete for [food](#)) I cannot find much on the feeding ecology of Plaice, but we have a relatively large sample size here, so I think the results are reliable.





Number Density Plot from Number Distribution



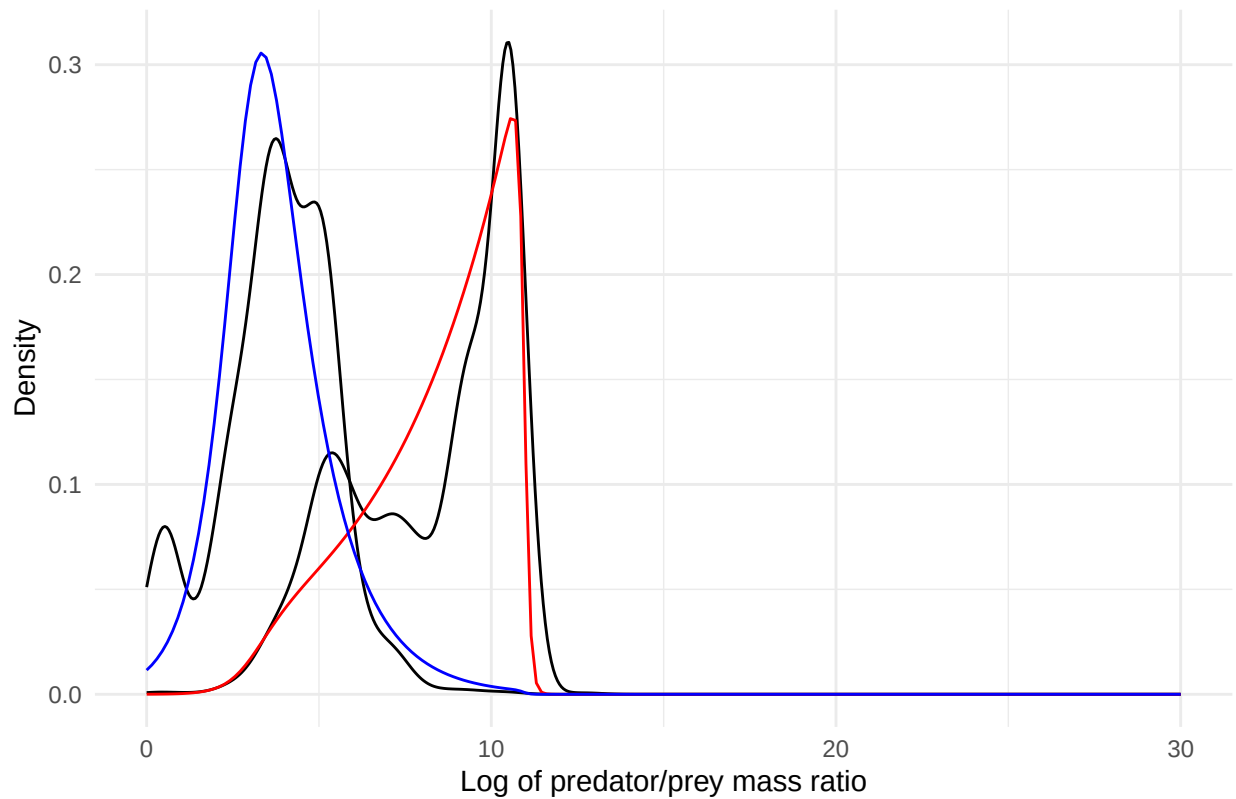


Poor Cod

[Poor Cod PDF file](#) [Poor Cod RMD file](#)

The fit for poor cod is okay. We do not have a large sample size here, which may be why there are many small peaks but also may be due to the seasonal, regional and ontogenic differences in diet [source](#) Regardless, the fit we have covers the range of PPMR in the number/biomass distributions. It jsut hides the feeding modes that may or may not be present.

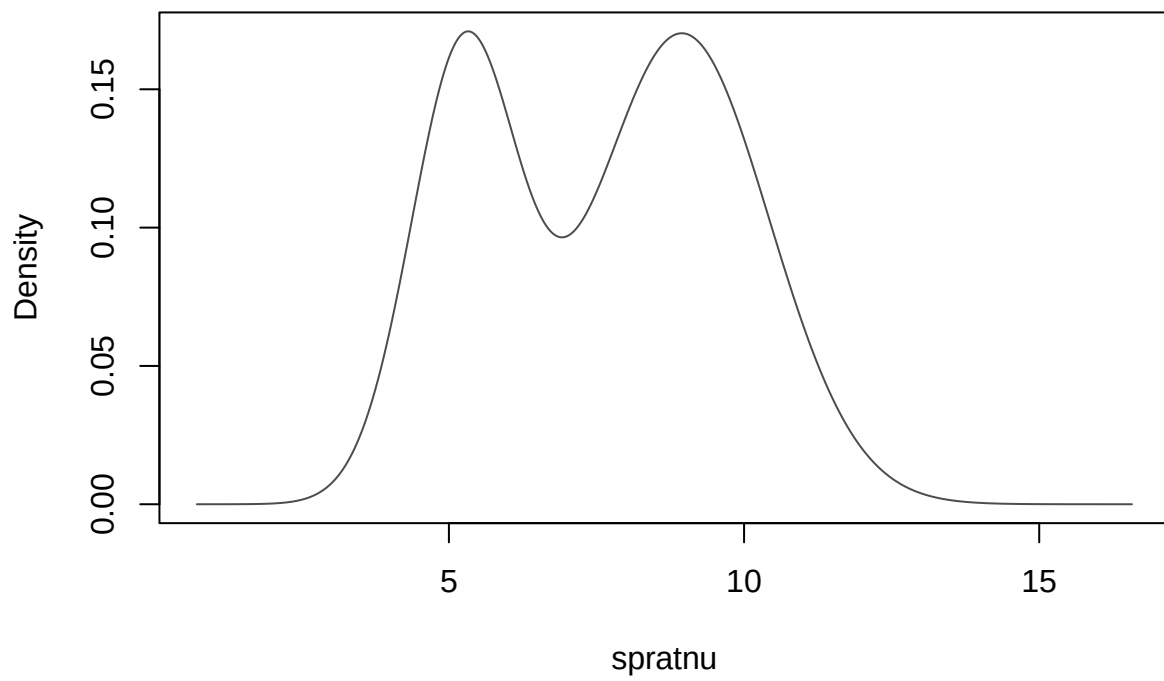
Poor Cod – Fitted to biomass, shift to number



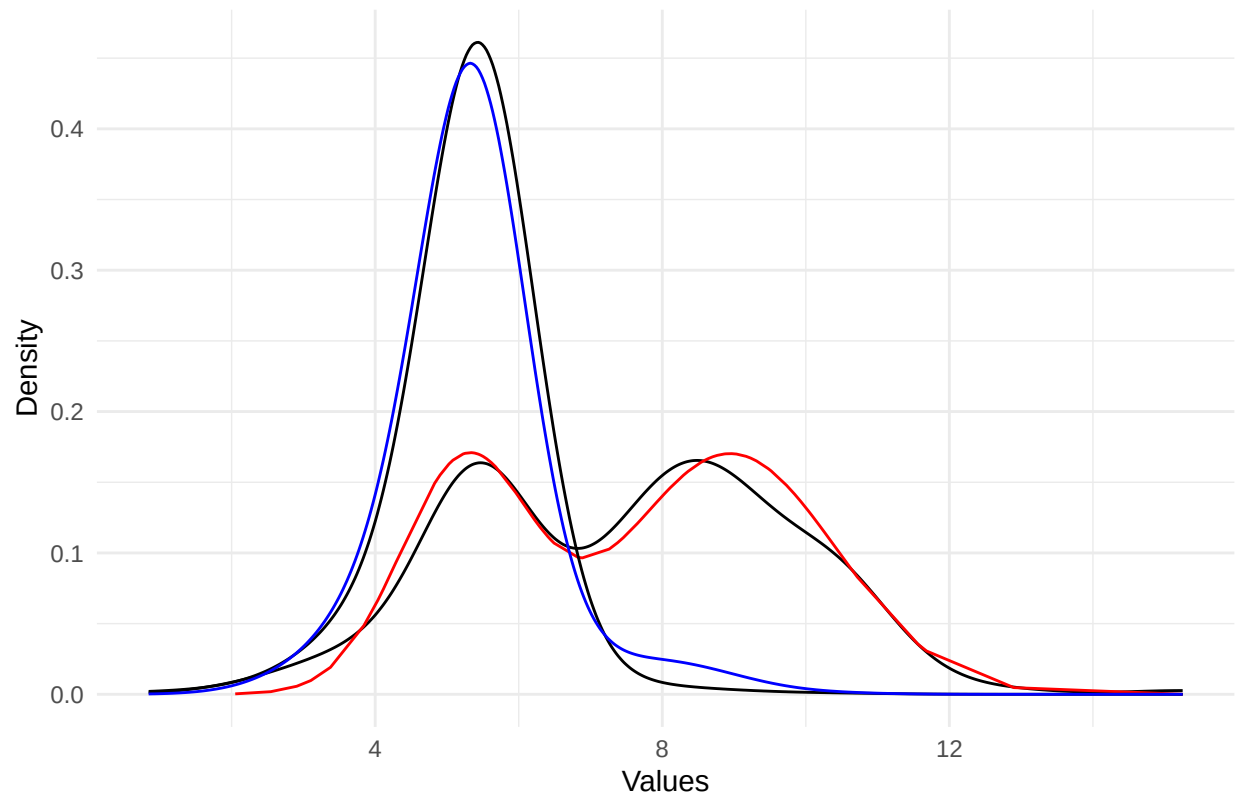
Sole

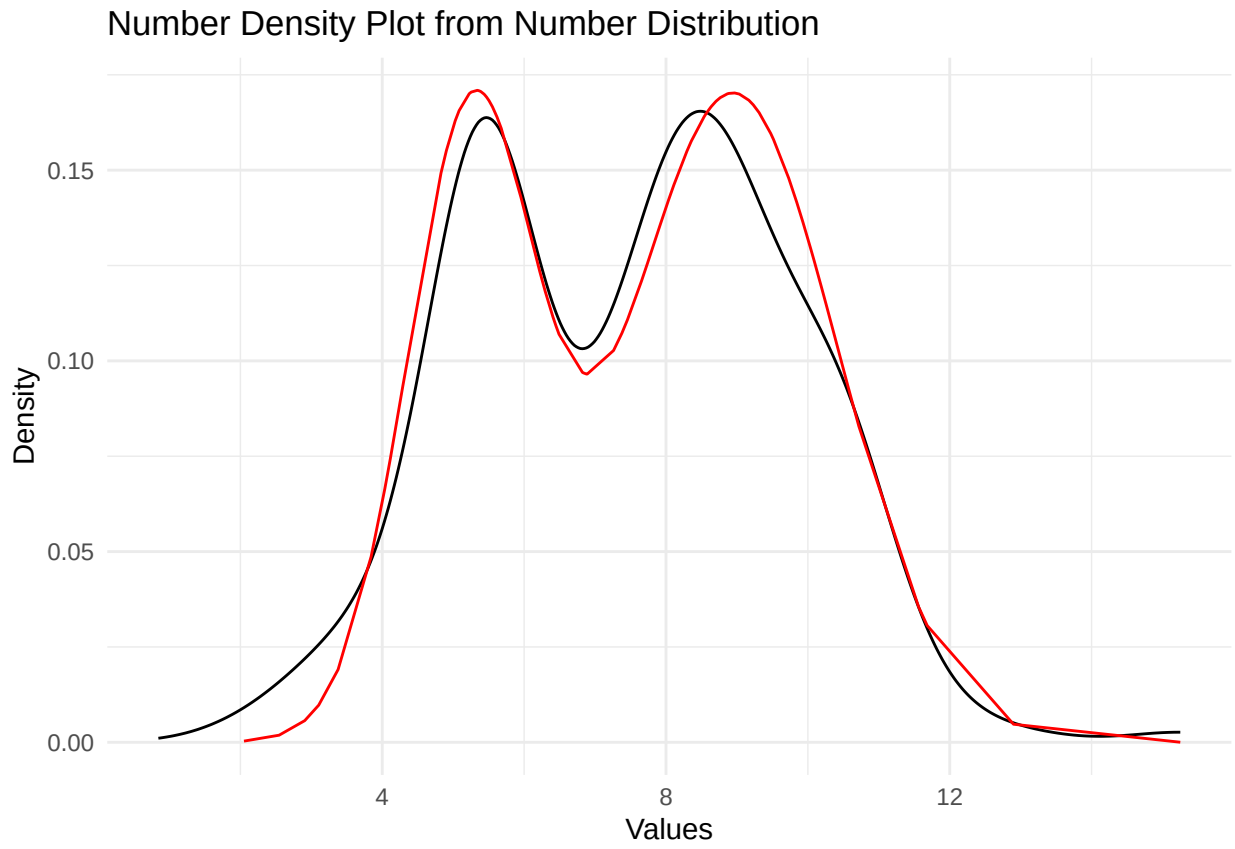
[Sole PDF file](#) [Sole RMD file](#)

Sole's fit is extremely good. But it has a very small sample size (229 not accounting for the weightings). There appears to be two different feeding modes, which is corroborated by the literature, as it has been shown that sole favours high energetic prey (high biomass content) when the female soles are approaching their reproductive cycle. [source](#) We can see this as the 2 gaussians fit to this data.

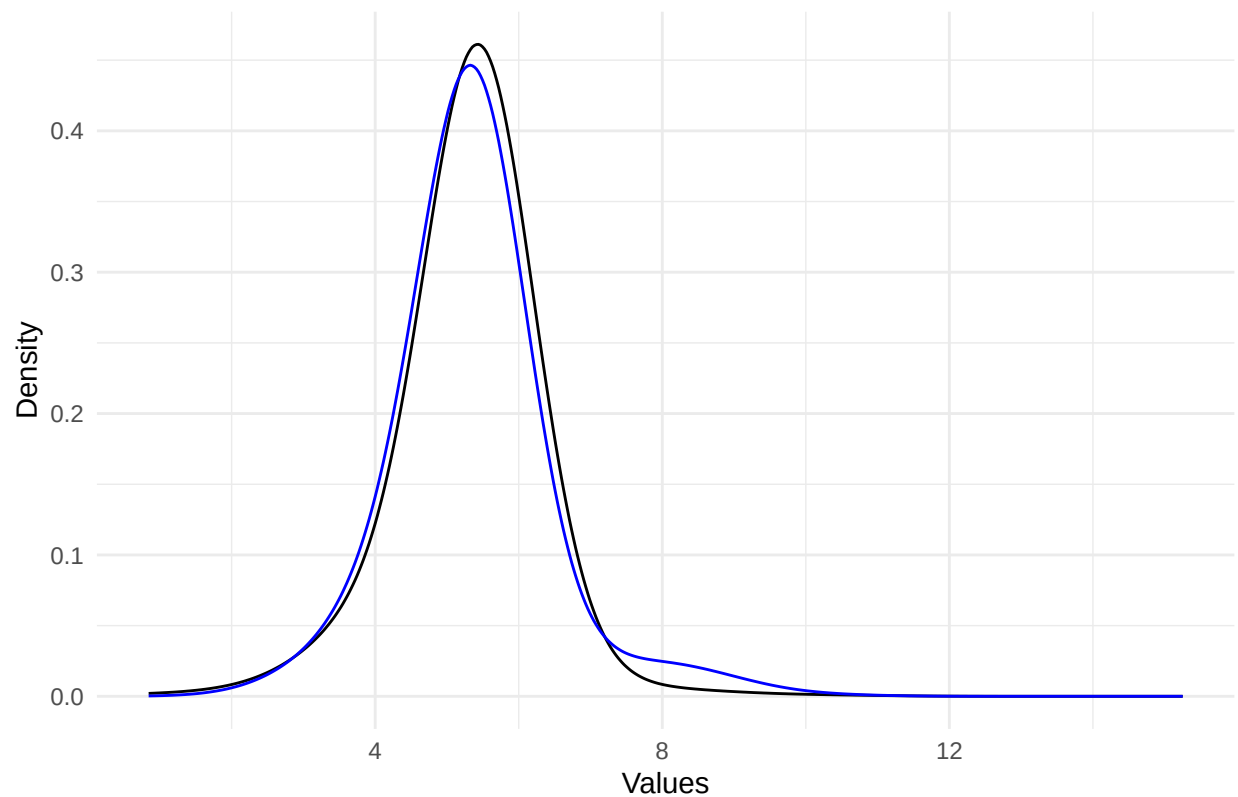


Sole – Fitted to number, shifted to biomass





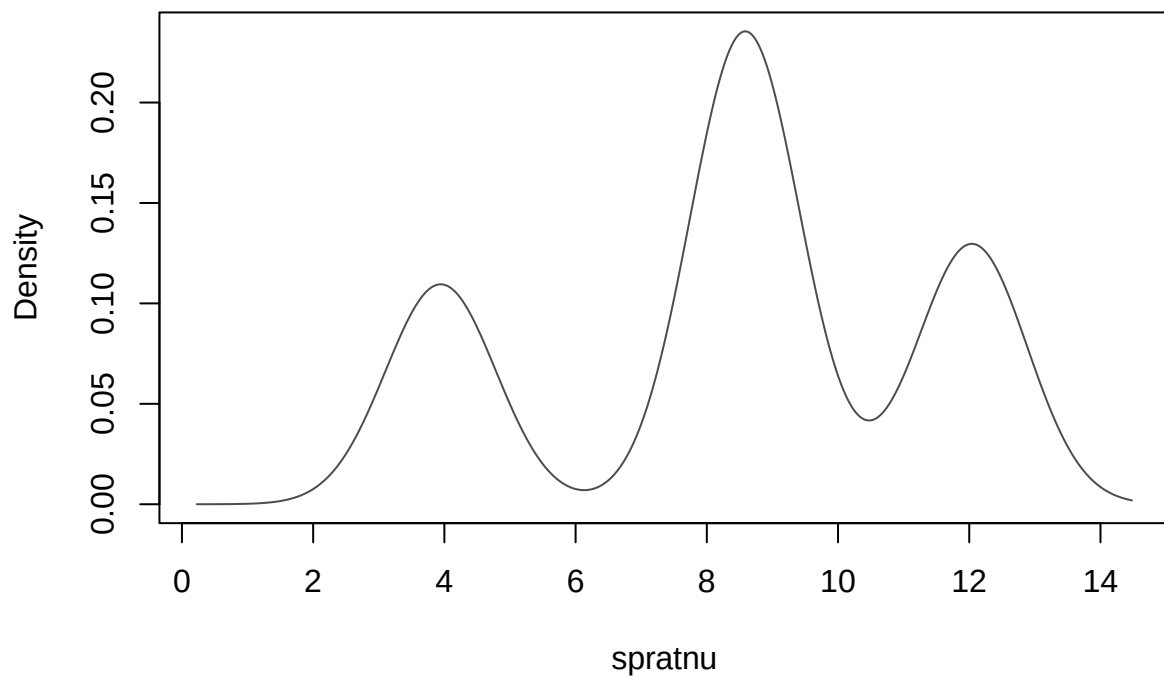
Diet Density Plot from Number Distribution



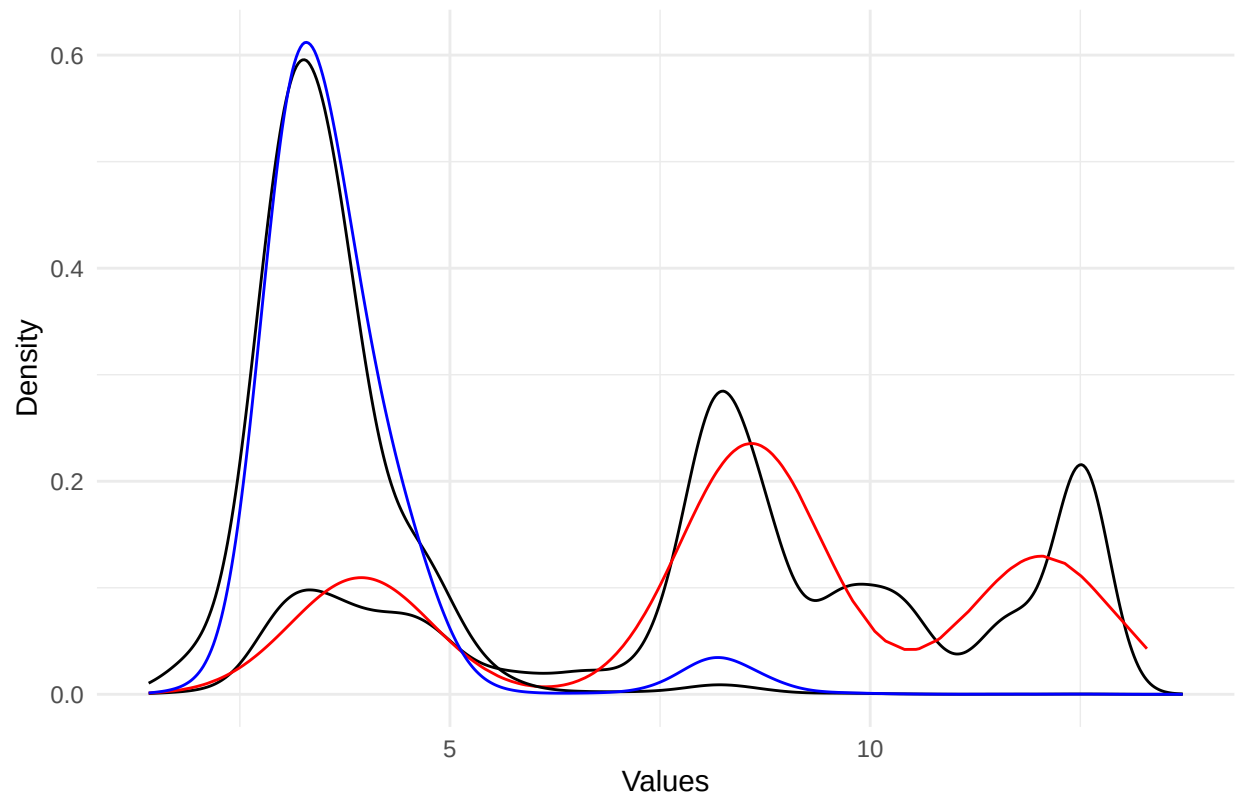
Sprat

[Sprat PDF file](#) [Sprat RMD file](#)

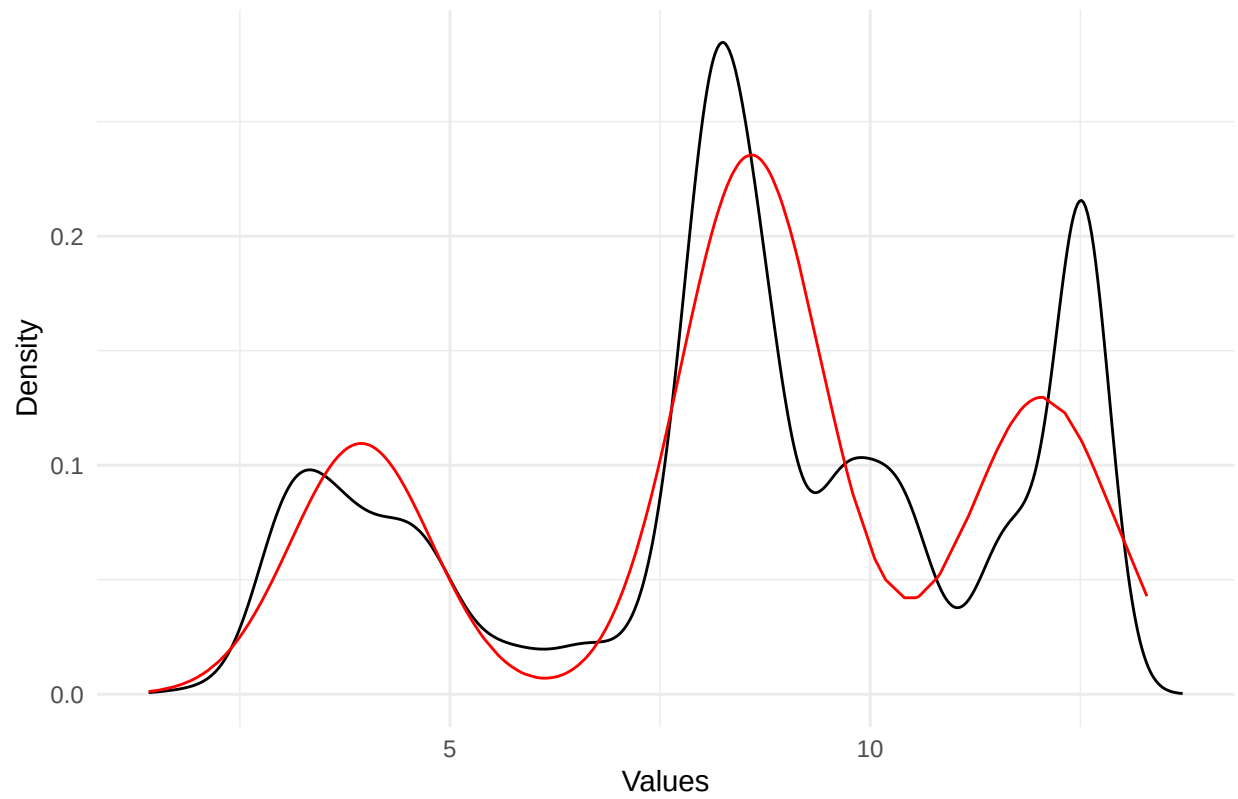
Sprat has a good fit, which is surprising given the PPMR distribution is complicated. There are a few different things each of the feeding modes could be, seasonal differences, day/night feeding changes, or maybe different regions. [Source](#) Regardless, the fit is very good, and it has a decent enough sample size to believe.

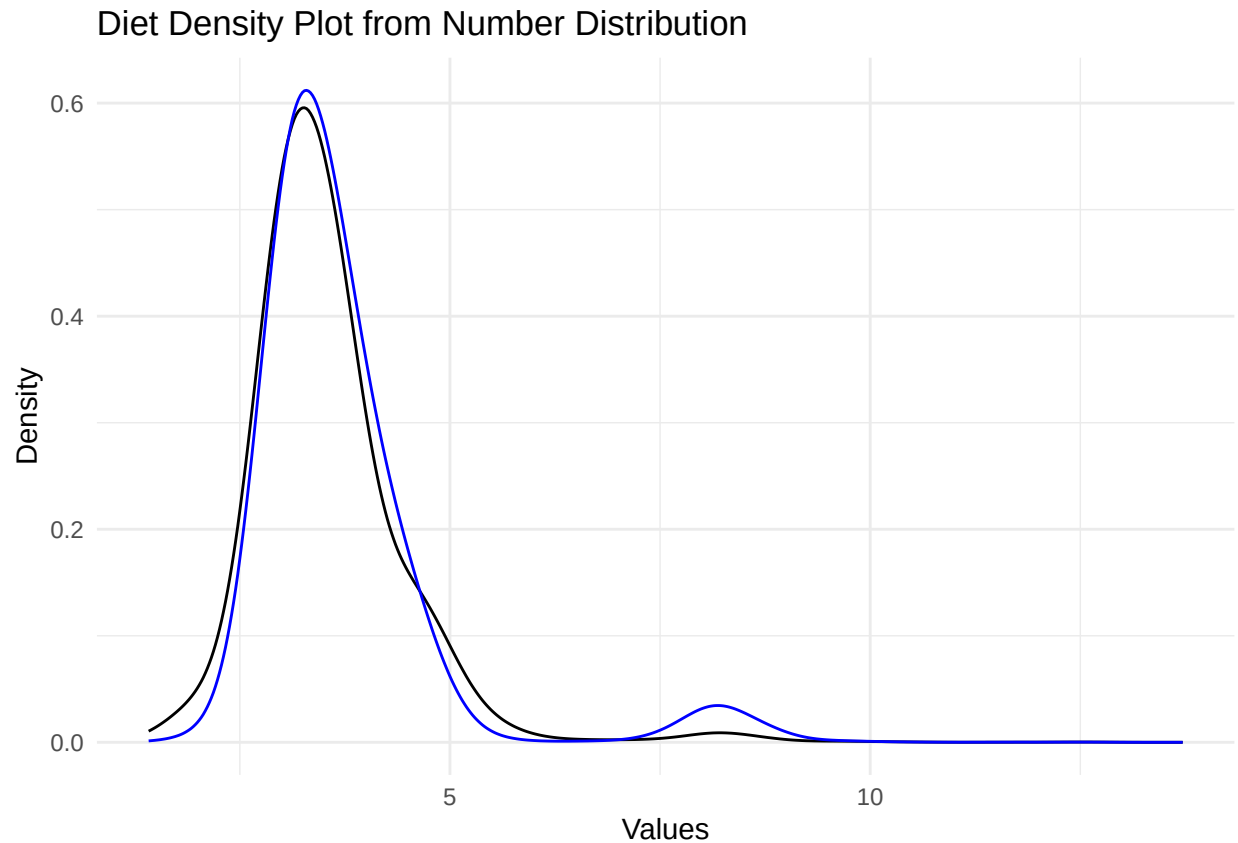


Sole – Fitted to number, shifted to biomass



Number Density Plot from Number Distribution

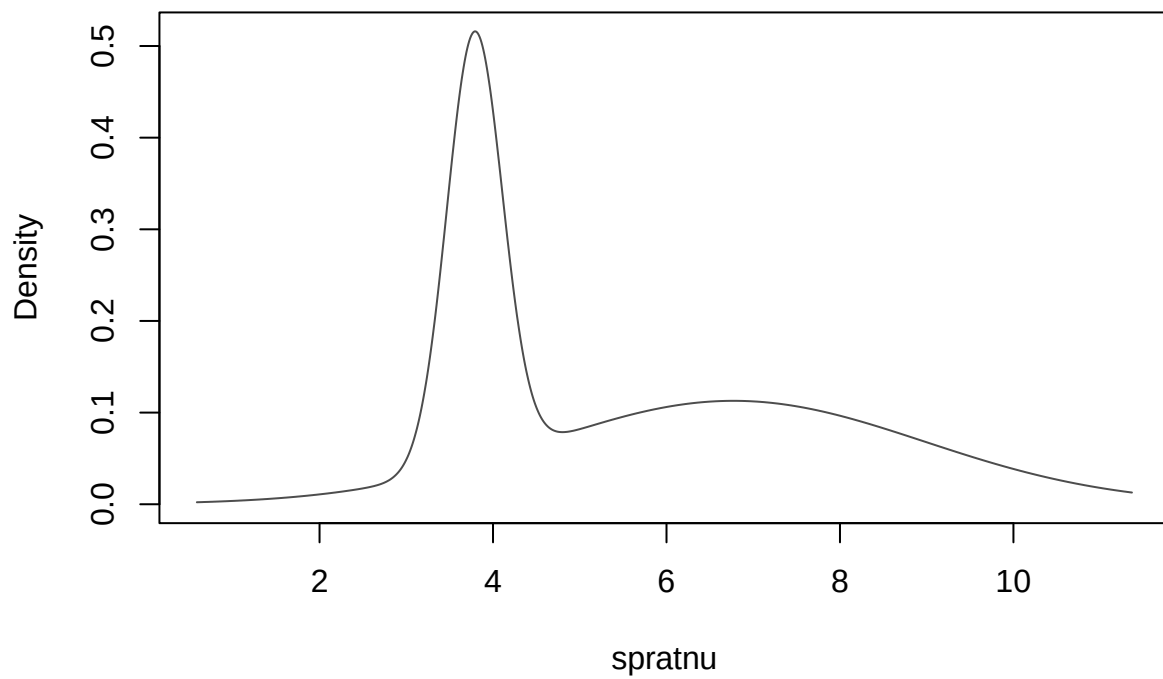


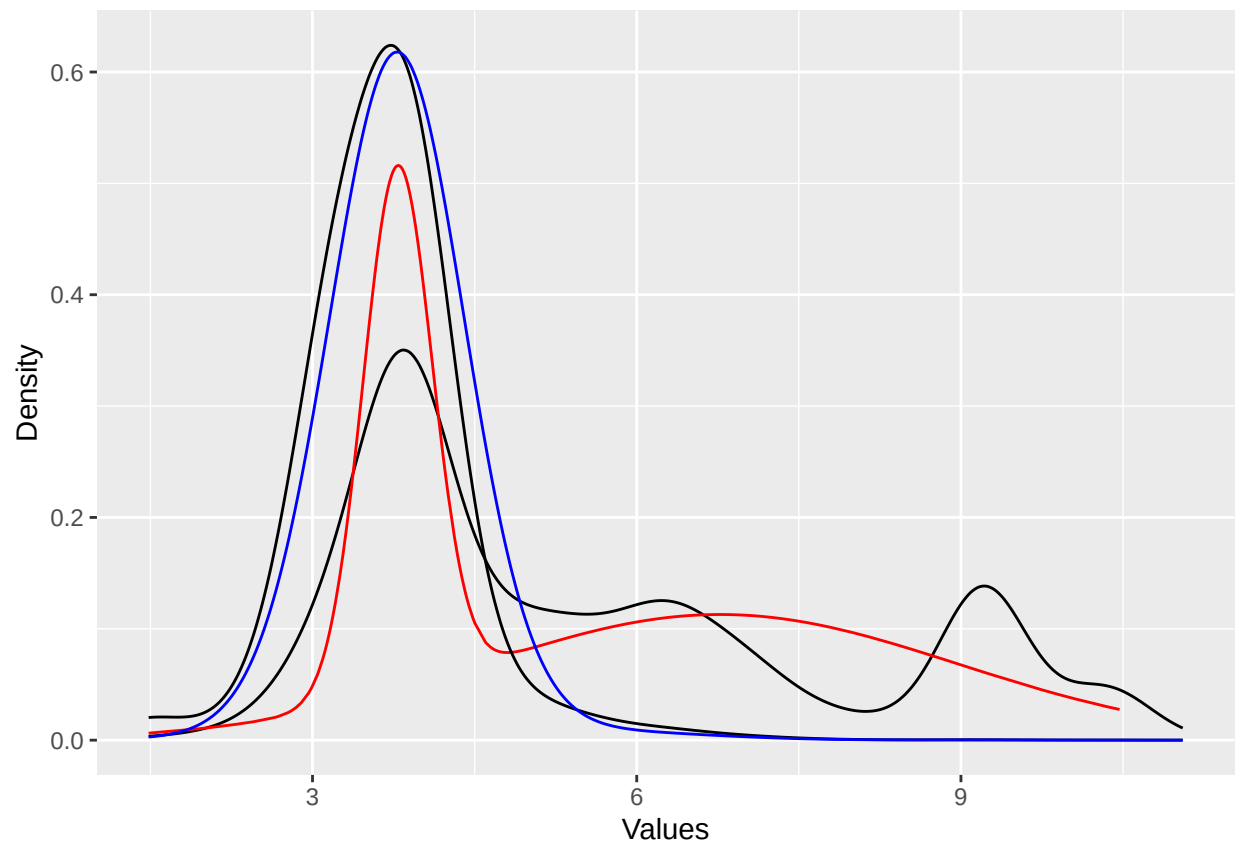


Spurdog

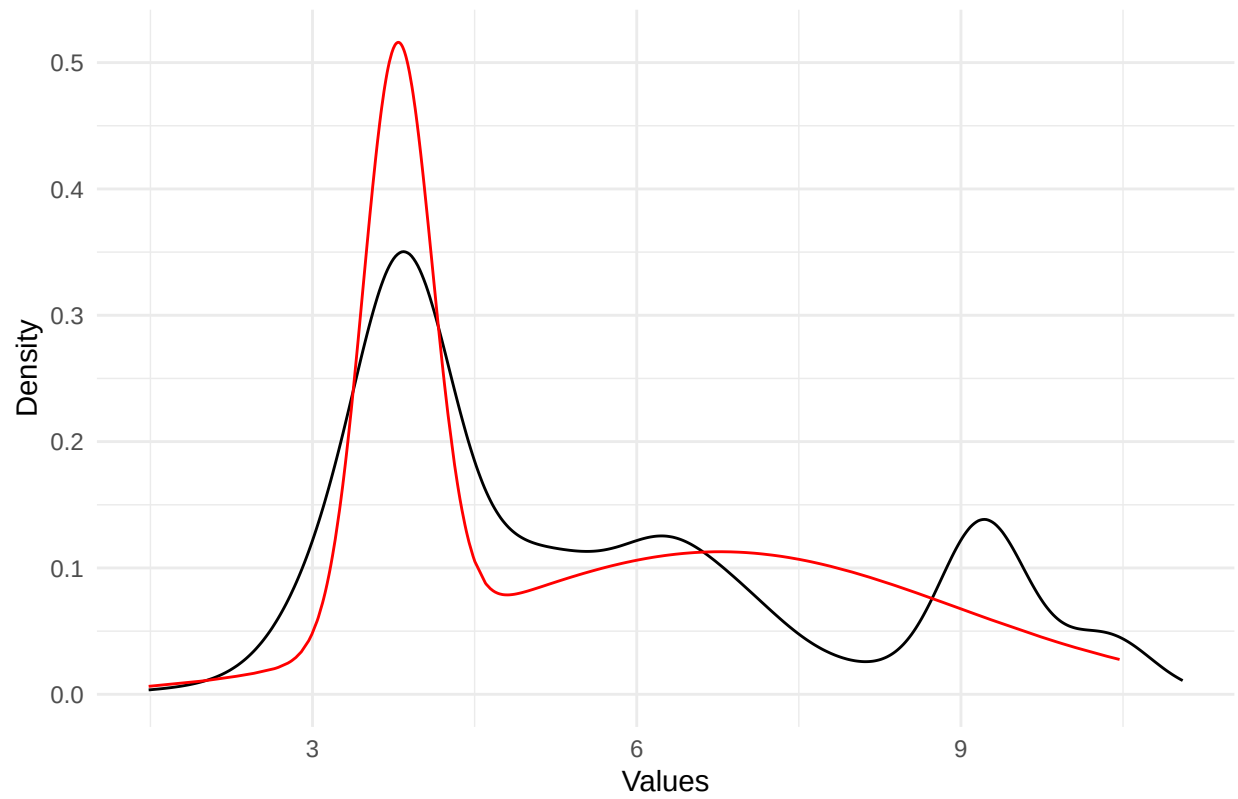
[Spurdog PDF file](#) [Spurdog RMD file](#)

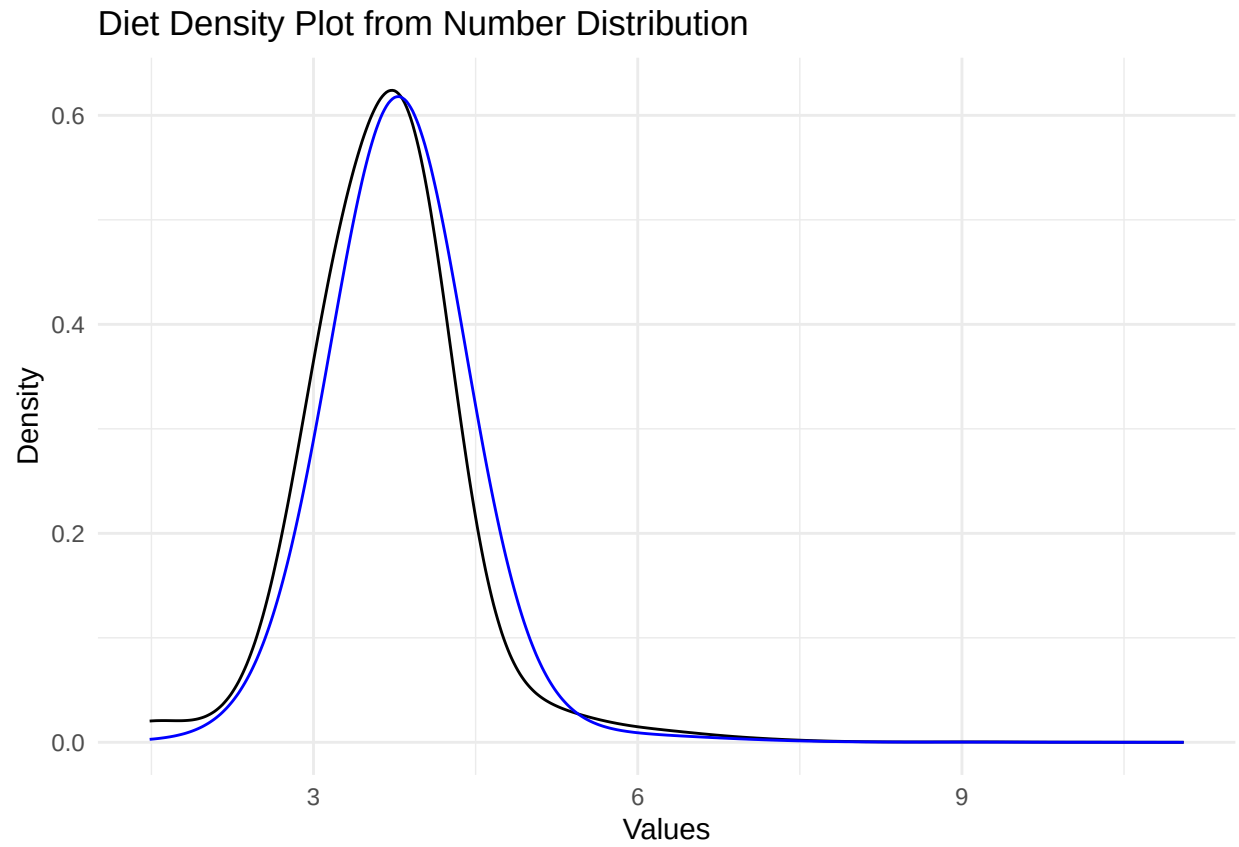
The Spurdog fit is okay, it is a difficult distribution to fit as it appears to have multiple feeding modes. Regional, seasonal and ontogenic diet differences were found [Source](#) so we could be observing this in the data. Additionally, the wide PPMR range is corroborated by [Source](#), and this source also states that teleosts = 80.32, cephalopods 11.96% and crustaceans 7.68%, which could be these 3 feeding modes that we are observing. Still, this is a good fit and describes our data.





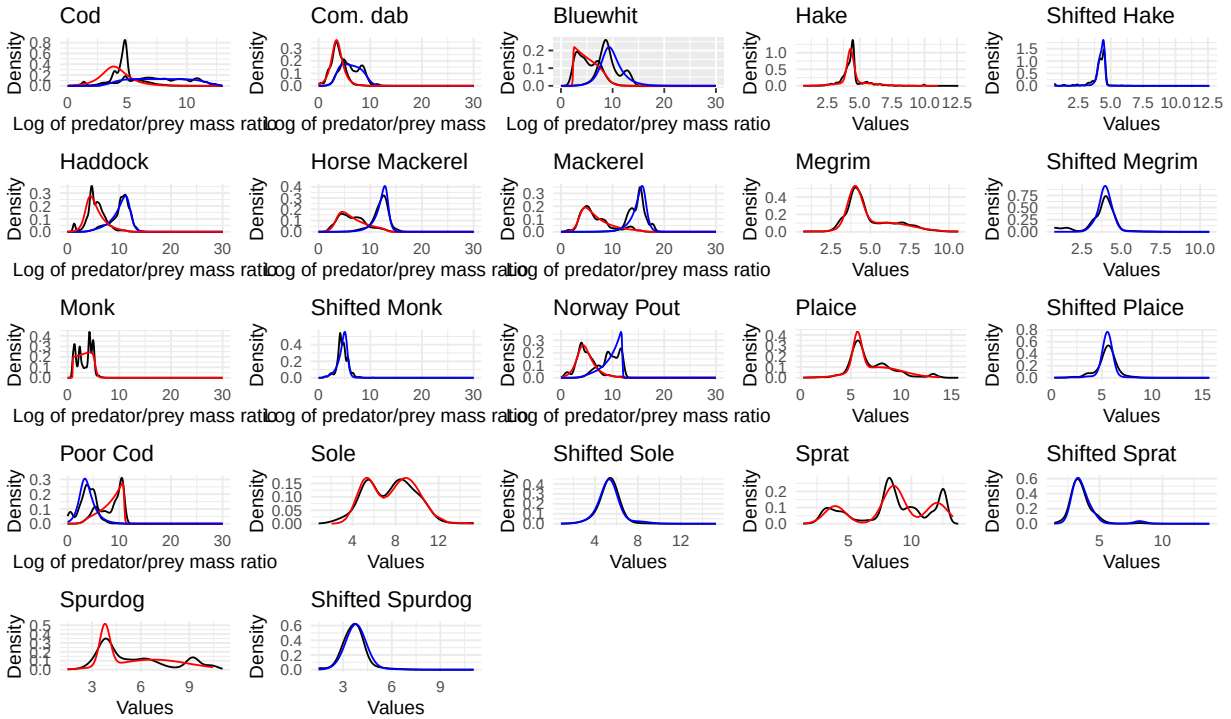
Number Density Plot from Number Distribution





Compilation plot

Some of the plots do not have the biomass and number densities plotted together, as they overlap.



Compiling the parameters

I compiled the parameters manually into the list, but here is the outputs from the code here.

And here is the code that puts it into a list

```
species_data <- list(
  "blue whiting" = list(
    fit = "ok",
    parameters = c(alpha = -0.1310297, ll = 2.3405346, ul = 22.9956791, lr =
      9.0047846, ur = 1.4277256)
  ),
  "whiting" = list(
    fit = "not good",
    parameters = c(alpha = -0.3145214, ll = 3.6304283, ul = 4.5452752, lr =
      13.3696362, ur = 1.6451461)
  ),
  "cod" = list(
    fit = "ok",
    parameters = c(alpha = -0.0005523561, ll = 3.7634229256, ul = 2.2665059865,
      lr = 11.7093958978, ur = 1.8644950972)
  ),
  "common dab" = list(
    fit = "very good",
    parameters = c(alpha = -0.05424663, ll = 3.39867067, ul = 2.29557044, lr =
      9.44078251, ur = 1.66081495)
  ),
  "european hake" = list(
    fit = "good",
    parameters = list(
```

```

    pro = c(0.6450399, 0.3549601),
    mean = c(4.286325, 4.710199),
    sigmasq = c(0.06341901, 2.44856948)
  )
),
"haddock" = list(
  fit = "good",
  parameters = c(alpha = 0.4805395, ll = 3.8865861, ul = 2.1827656, lr =
    11.7890770, ur = 2.3939047)
),
"herring" = list(
  fit = "no fit",
  parameters = NA
),
"Horse Mackerel" = list(
  fit = "good",
  parameters = c(alpha = -0.2374178, ll = 3.5785207, ul = 1.9266416, lr =
    13.1902886, ur = 3.3245700)
),
"Mackerel" = list(
  fit = "very good",
  parameters = c(alpha = -0.2856444, ll = 3.7936324, ul = 2.0442341, lr =
    16.1008630, ur = 2.7353730)
),
"Megrim" = list(
  fit = "ok",
  parameters = list(
    pro = c(0.5504123, 0.4495877),
    mean = c(4.015029, 5.947145),
    sigmasq = c(0.2139633, 2.8654756)
  )
),
"monkfish" = list(
  fit = "not good",
  parameters = c(alpha = 0.0700759, ll = 0.9805321, ul = 54.0317054, lr =
    5.3291042, ur = 4.3538944)
),
"norway pout" = list(
  fit = "ok",
  parameters = c(alpha = -0.6399368, ll = 4.0059442, ul = 1.7483370, lr =
    11.8681193, ur = 13.6461300)
),
"plaice" = list(
  fit = "ok",
  parameters = list(
    pro = c(0.4013671, 0.5986329),
    mean = c(5.650236, 7.762250),
    sigmasq = c(0.1959833, 5.8912814)
  )
),
"poor cod" = list(
  fit = "good",
  parameters = c(alpha = 0.03769714, ll = 3.89995523, ul = 1.84539297, lr =

```

```

11.07162811, ur = 9.65116209)
),
"sole" = list(
  fit = "very good",
  parameters = list(
    pro = c(0.370072, 0.629928),
    mean = c(5.256386, 8.947248),
    sigmasq = c(0.820242, 2.179852)
  )
),
"Sprat" = list(
  fit = "ok",
  parameters = list(
    pro = c(0.2307388, 0.4959397, 0.2733215),
    mean = c(3.943746, 8.584915, 12.039642),
    sigmasq = 0.7068256
  )
),
"spurdog" = list(
  fit = "not good",
  parameters = list(
    pro = c(0.377416, 0.622584),
    mean = c(3.787757, 6.772024),
    sigmasq = c(0.1022199, 4.8450106)
  )
)
)

```